

SDR Against Smart TVs; URL and channel injection attacks



DefCON 27, Pedro Cabrera

@PCabreraCamara



About Me



Industrial engineer, UAV professional pilot, Radio Ham (EA4HCF)

Ethon Shield, Founder

2018 RSA: *"Parrot Drones Hijacking"*

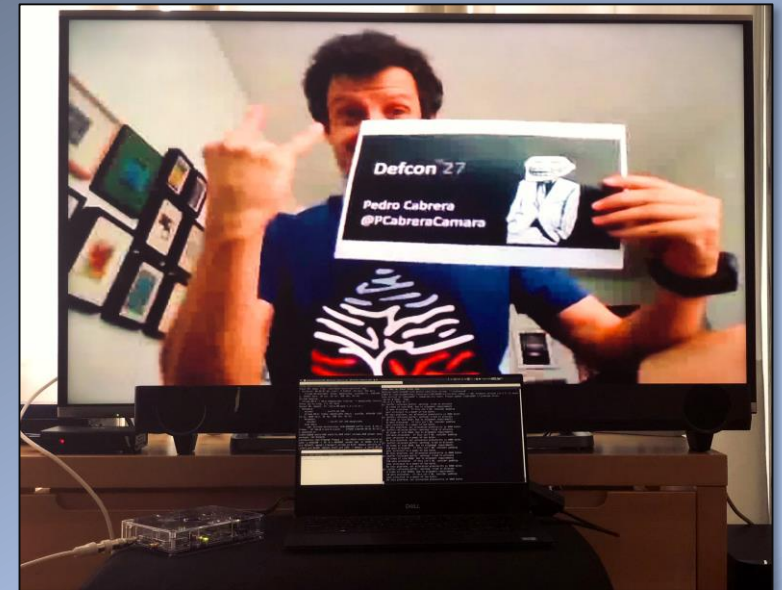
2017 BH Asia Trainings (+ Simon Roses):

"Attacking 2G/3G mobile networks, smartphones and apps"

RogueBTS, IMSICatchers, FakeStations: www.fakebts.com



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DefCON 27, Pedro Cabrera



This.Presentation:



- I. HbbTV 101. Digital TV introduction
- II. Hacking TV & HbbTV.
- III. HbbTV RX stations.
- IV. Targeting the Smart TV browser.
- V. Conclusions



[I] Hybrid Broadcast Broadband Television



The HbbTV specification extends DVB-T by introducing additional metadata formats that mix broadband Internet content into the digital television channel.

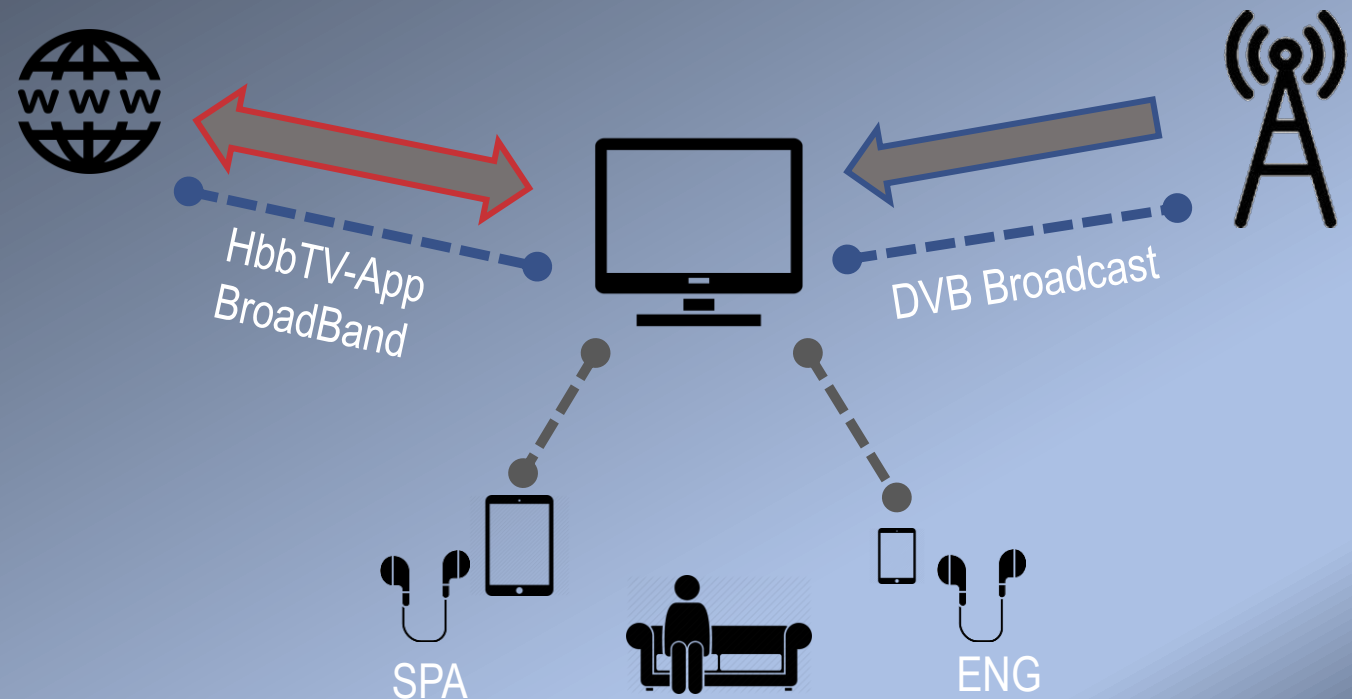
H4TV
project 



HbbTV
(2009)

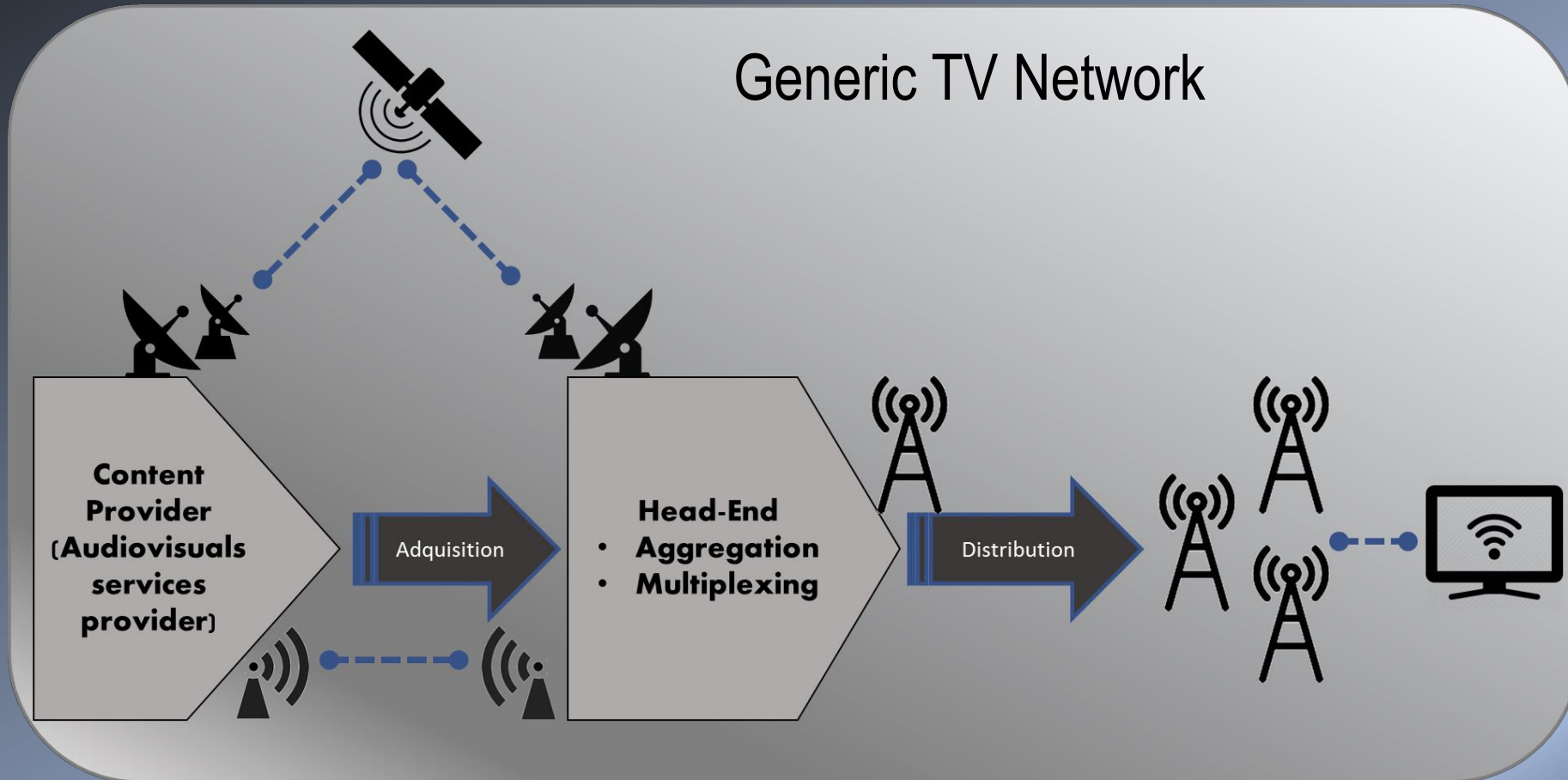
HTML profit
project 

Hybrid television because it
merges digital television
content and web content.





[I] TV Distribution Network



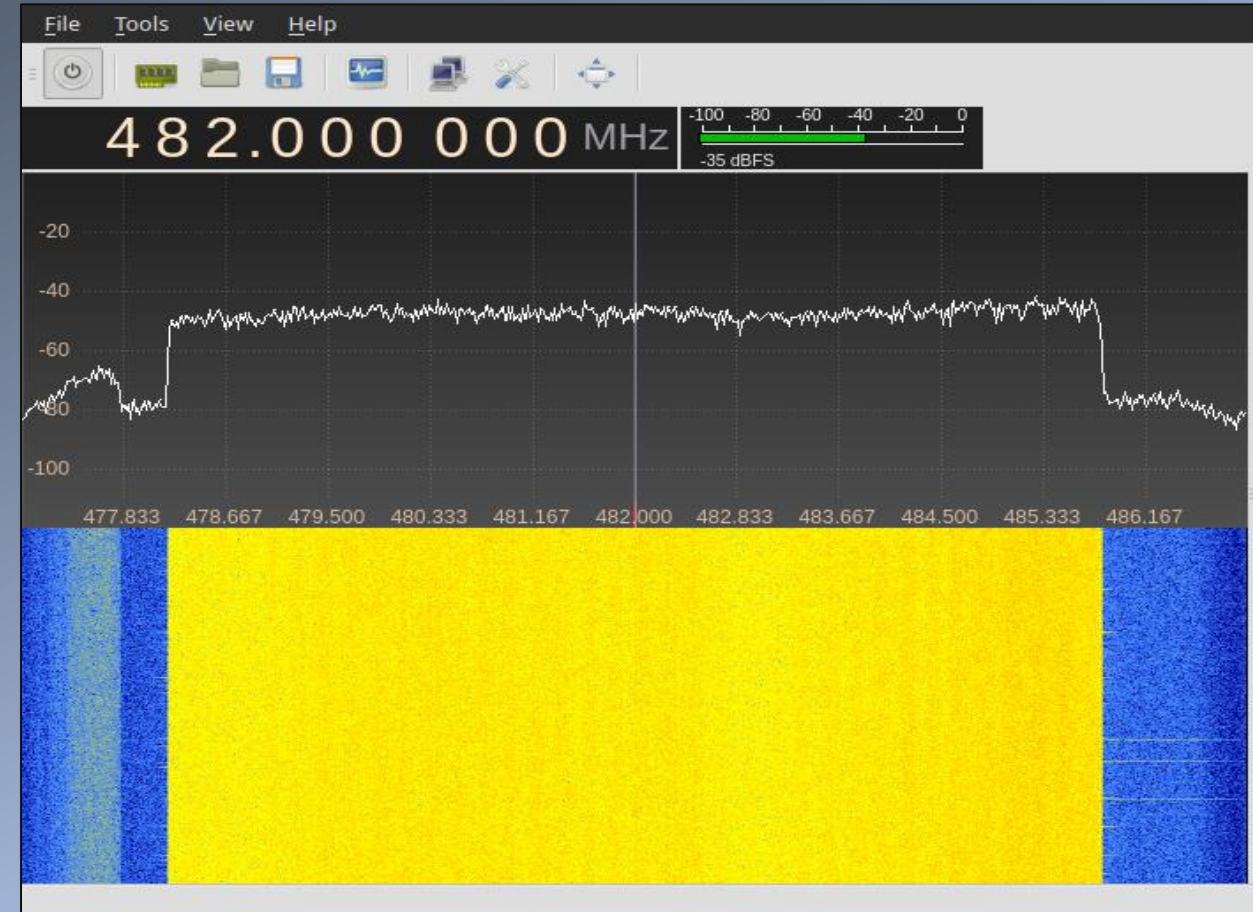


[I] DVB-T



DVB-T characteristics (Spain):

- 8 MHz bandwidth
- Transmission mode: 8k (6,817 carriers).
- Modulation schemes: 64 Quadrature Amplitude Modulation (OFDM)
- Code Rate for internal error protection: 2/3.
- Length of guard interval: 1/4.





[I] DVB-T



WHAT THE FUCK

AM I LOOKING AT

(Spain):

- 8 MHz bandwidth
- Transmission mode: 8k (6,817 carriers).
- Modulation schemes: 64 Quadrature Amplitude



Digit
Broad



rtlsdr sample rate



All Images Videos Shopping News More Settings Tools

About 37,100 results (0.48 seconds)

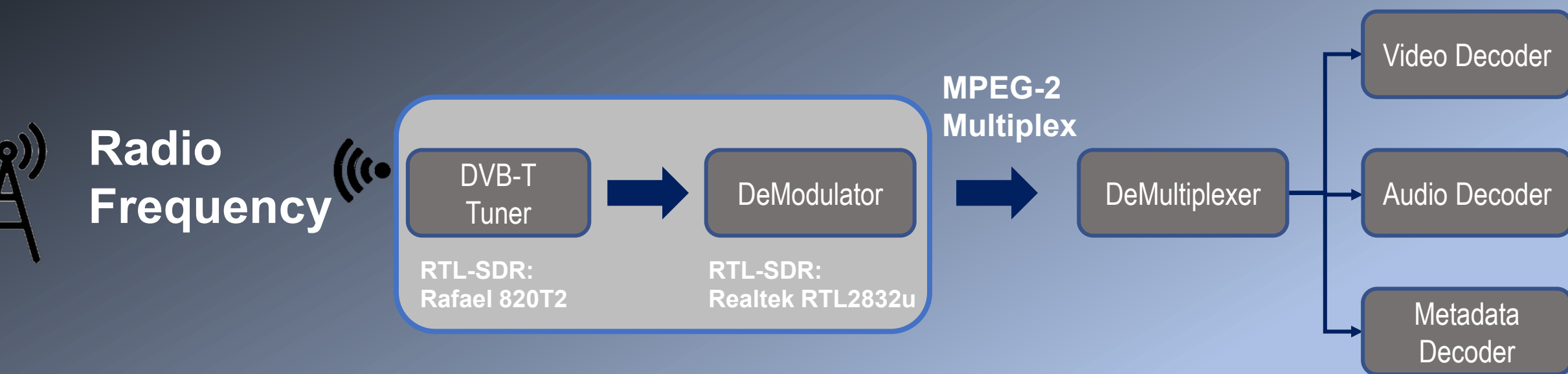
The maximum sample rate is **3.2 MS/s** (mega samples per second). However, the RTL-SDR is unstable at this rate and may drop samples. The maximum sample rate that does not drop samples is **2.4 MS/s**, however some people have had luck with 2.8MS/s and **3.2 MS/s** working well on some USB 3.0 ports.

About RTL-SDR - rtl-sdr.com
<https://www.rtl-sdr.com/about-rtl-sdr/>





[I] Generic DVB-T receiver

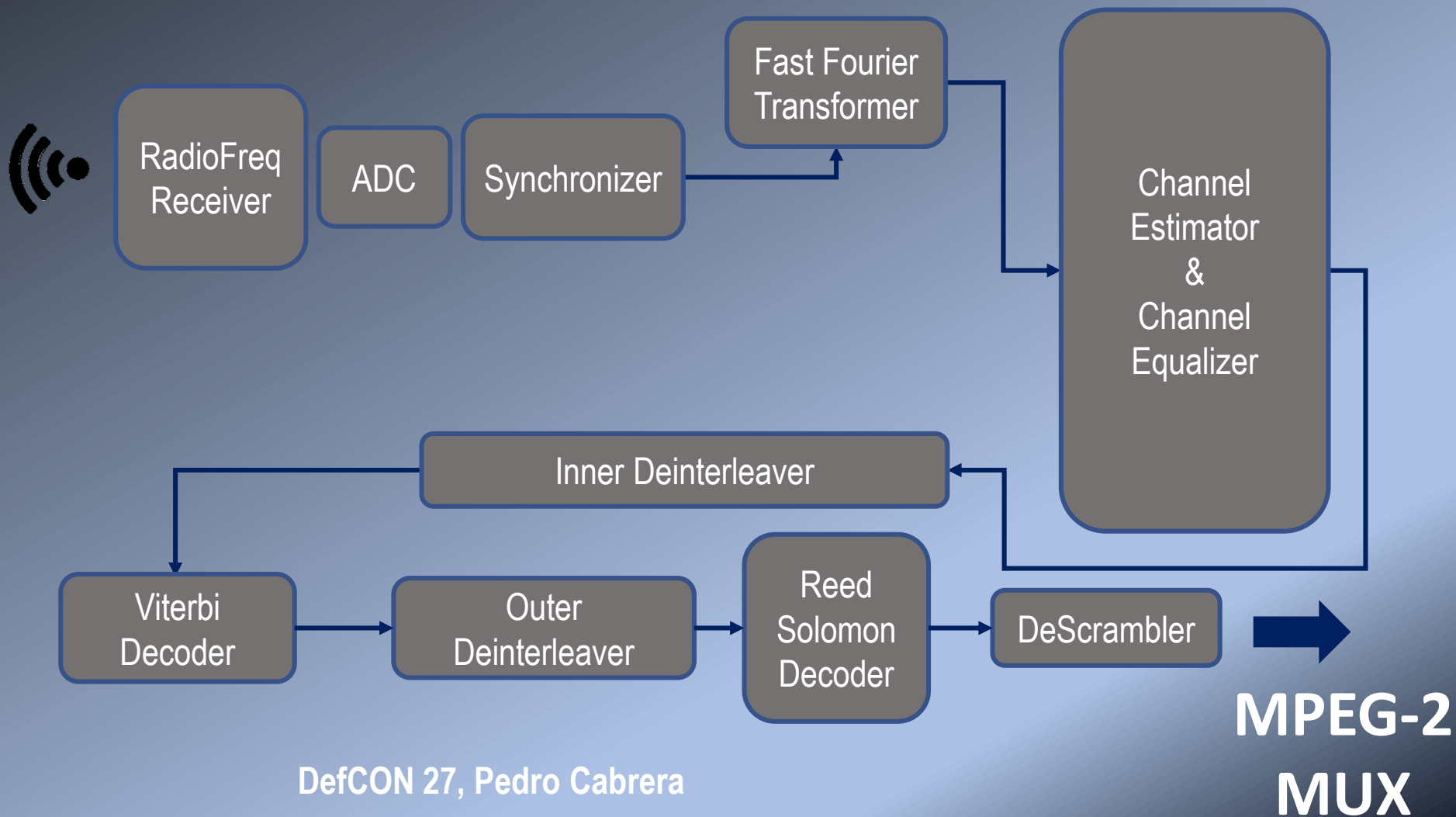




[I] DVB-T demodulator



**Radio
Frequency**





[I] DVB-T linux demodulator



- Bogdan Diaconescu (YO3IU)

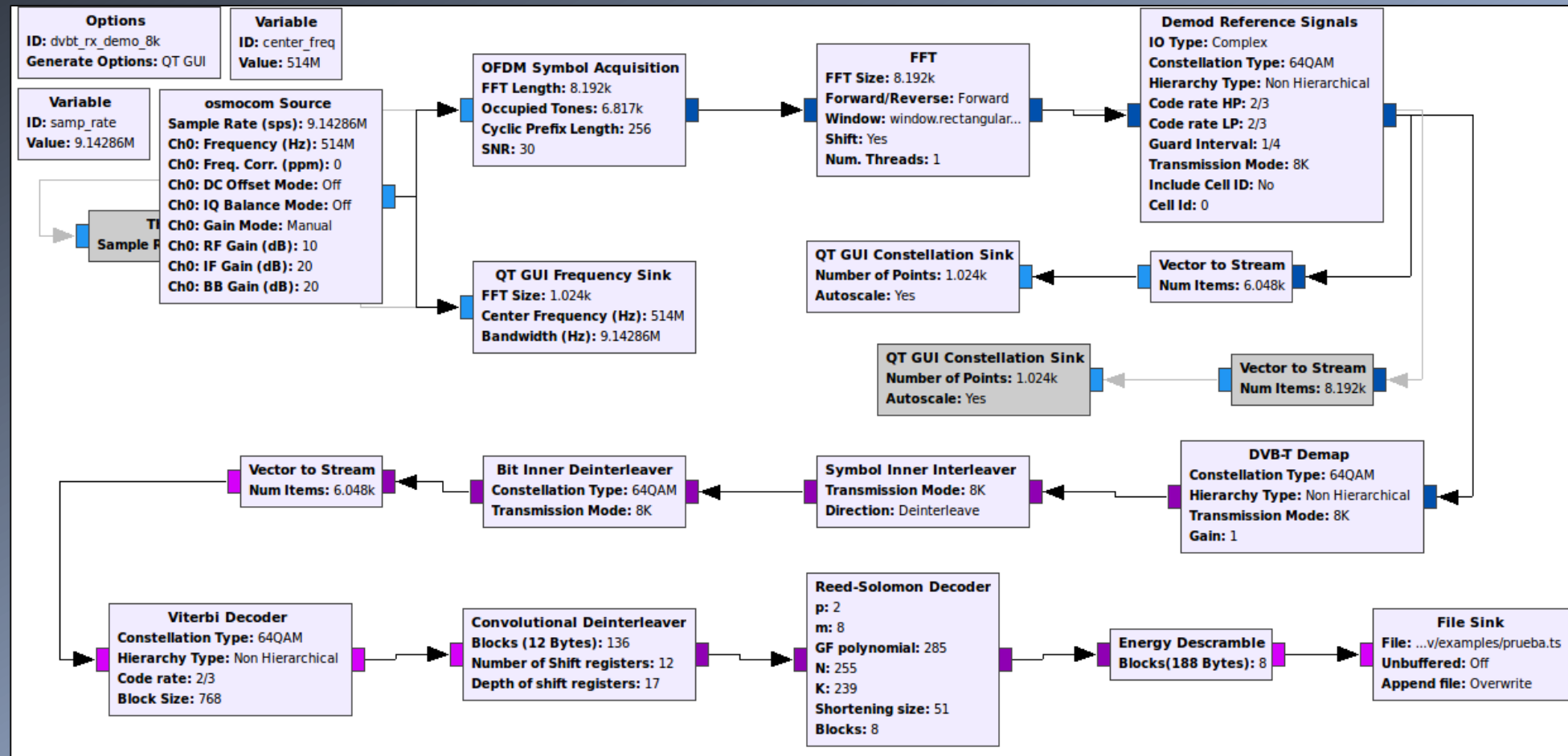
gr-dvbt
(USRP N210)

- GNU Radio:

gr-dtv (USRP)

- Ron Economos (W6RZ)

dtv-utils (BladeRF)





[I] DVB-T linux demodulator



DVB-T Modulation:

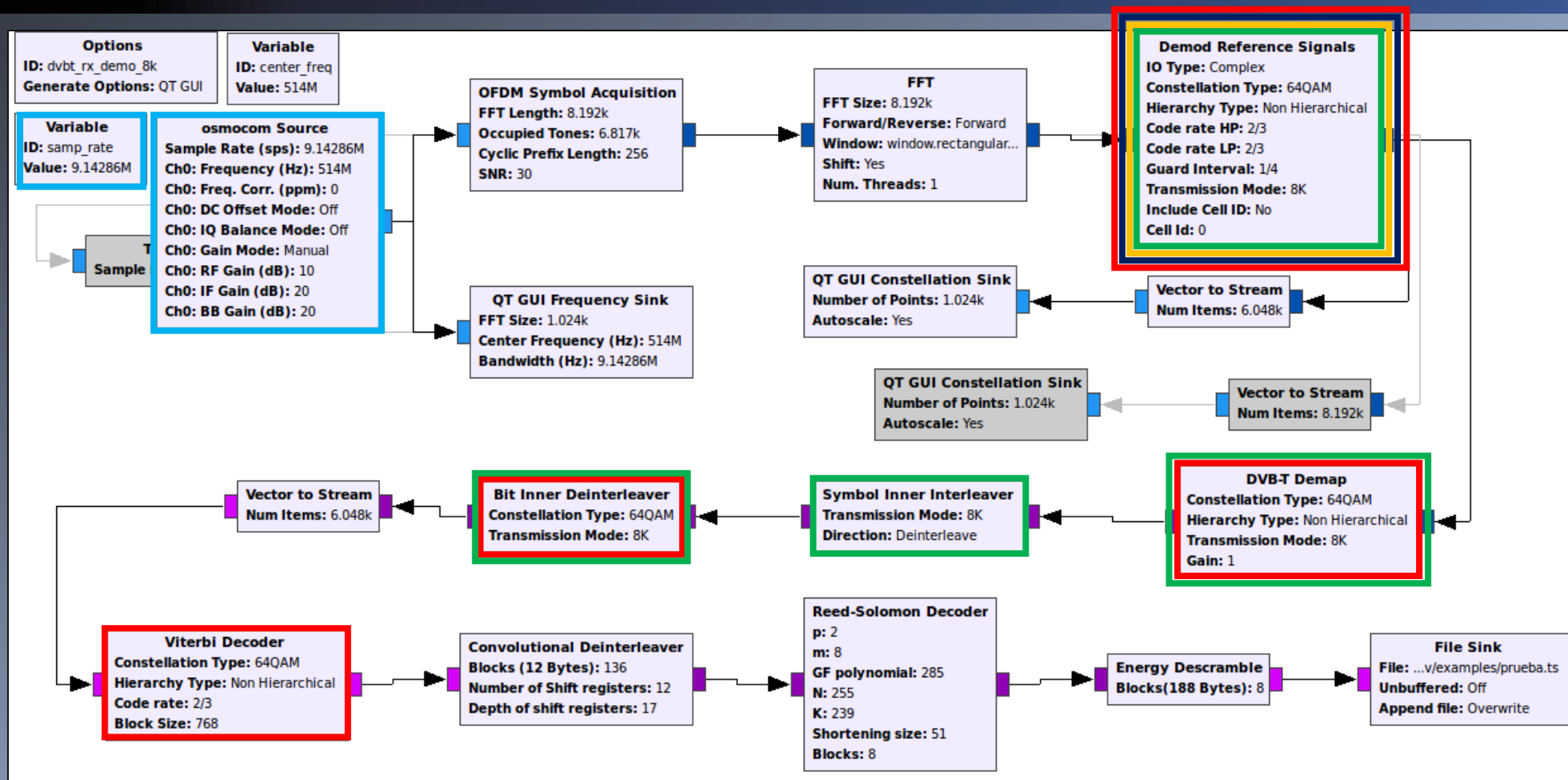
8 MHz Bandwidth (SR)

Transmission mode: 8k

Modulation scheme:
64 QAM

Code Rate: 2/3.

Length of guard interval:
1/4

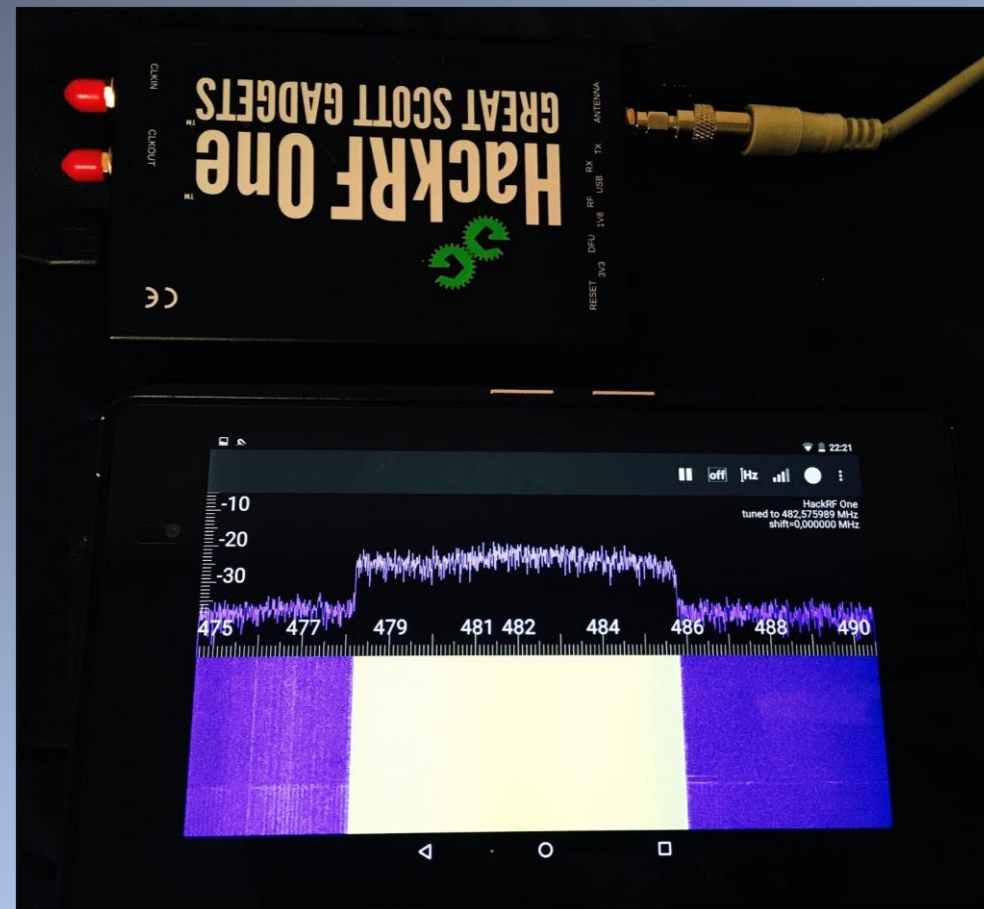


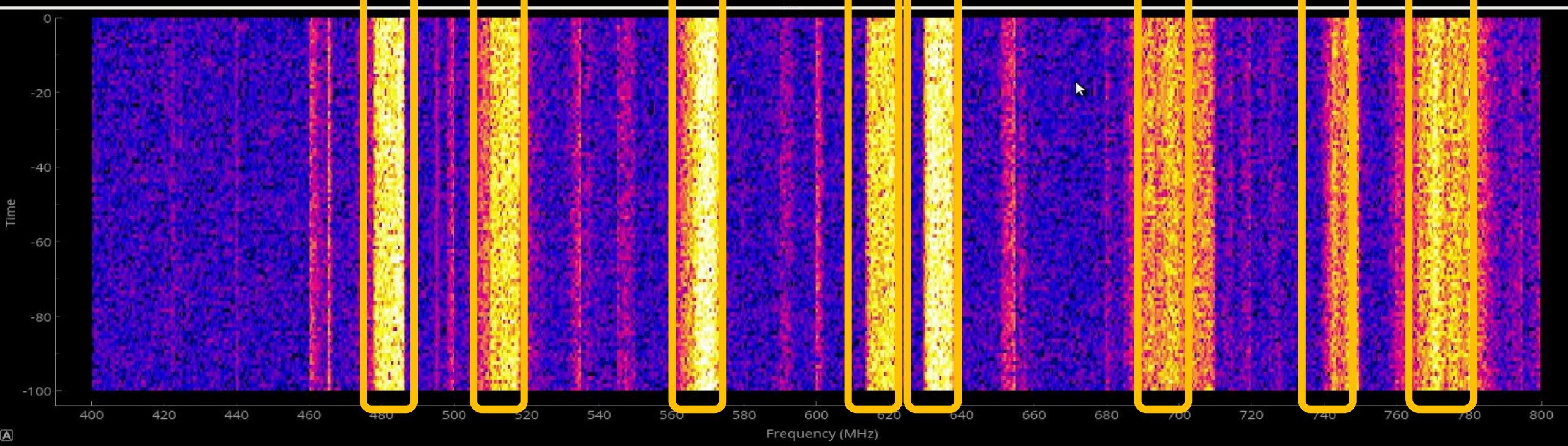
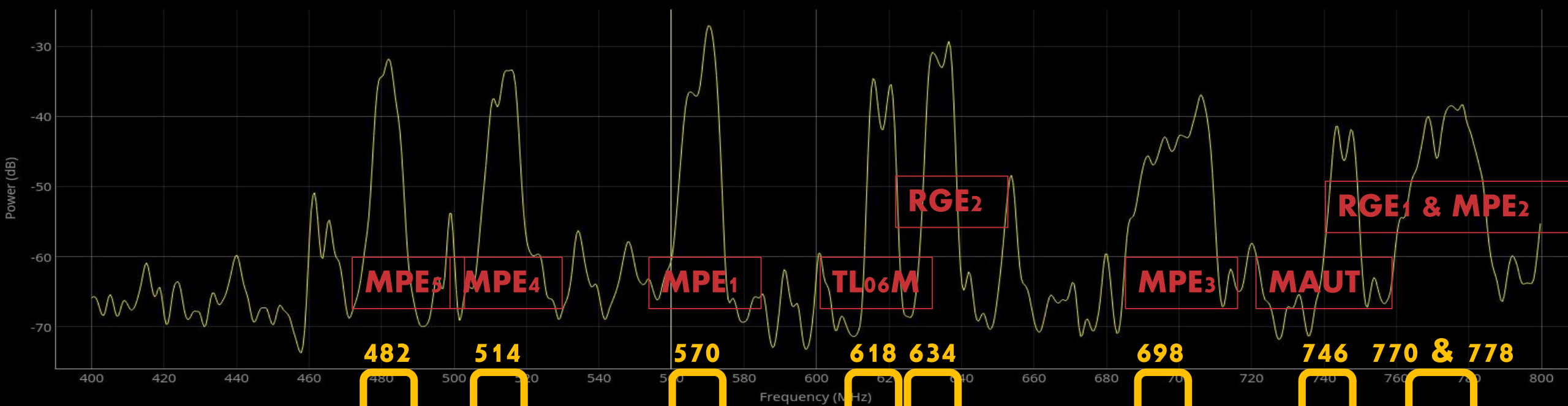


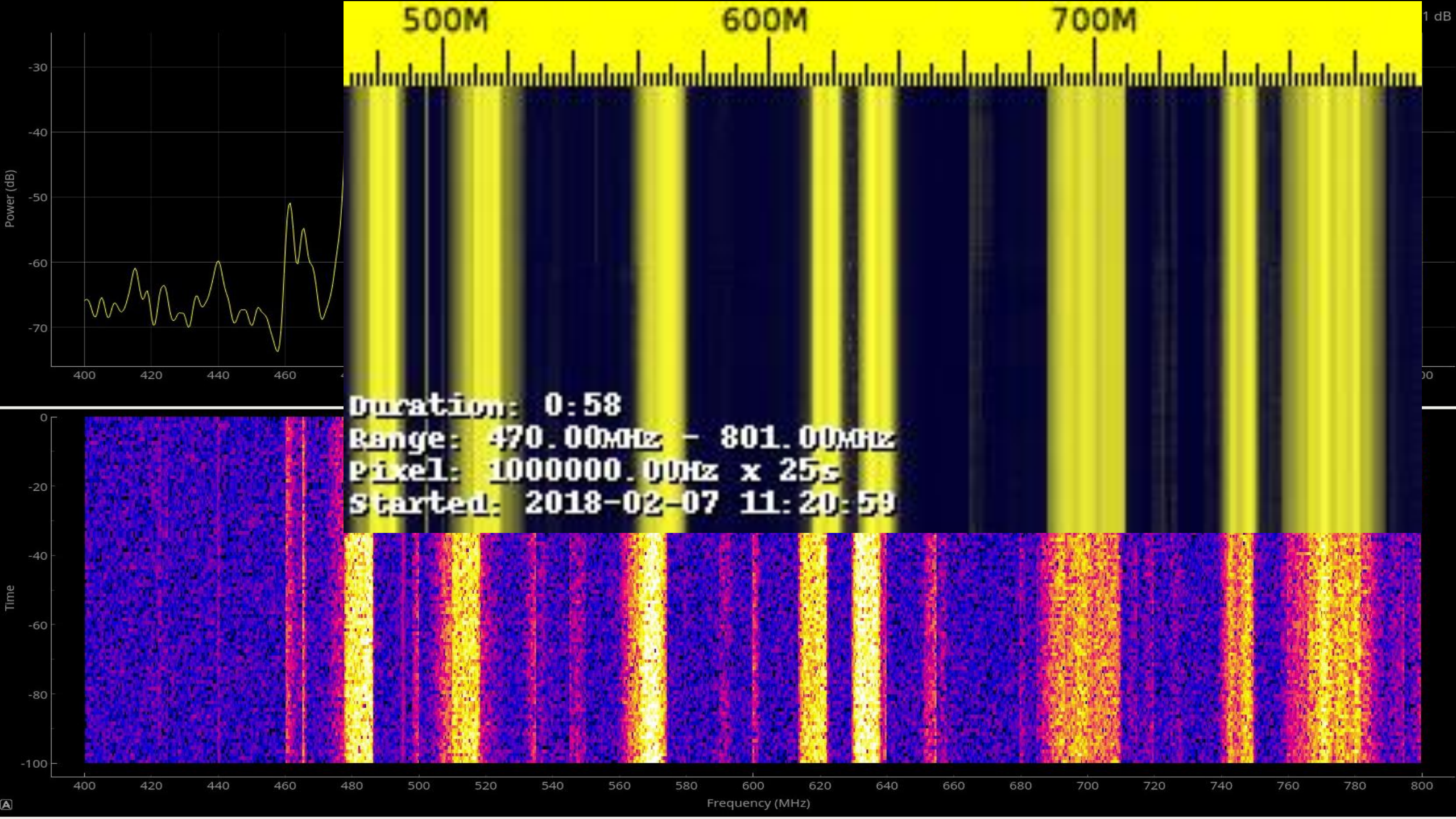
[I] TV Channels & Frequencies



Digital M ultiplex	Channel	Frequency	Digital M ultiplex	Channel	Frequency
M P E₅	atreseries HD	482.000.000	M P E₃	Telecinco	698.000.000
M P E₅	BeMad tv HD	482.000.000	M P E₃	Telecinco HD	698.000.000
M P E₅	Realmadrid TV HD	482.000.000	M P E₃	Cuatro	698.000.000
M P E₄	TRECE	514.000.000	M P E₃	Cuatro HD	698.000.000
M P E₄	Energy	514.000.000	M P E₃	FDF	698.000.000
M P E₄	mega	514.000.000	M P E₃	Divinity	698.000.000
M P E₄	Boing	514.000.000	MAUT	Telemadrid HD	746.000.000
M P E₁	PARAMOUNT CHANNEL	570.000.000	MAUT	Telemadrid	746.000.000
M P E₁	GOL	570.000.000	MAUT	LA OTRA	746.000.000
M P E₁	DMAX	570.000.000	MAUT	BOM	746.000.000
M P E₁	Disney Channel	570.000.000	RGE₁	La 2 HD	770.000.000
TL06M	TRECE	618.000.000	RGE₁	La 2	770.000.000
TL06M	Intereconomía TV	618.000.000	RGE₁	La 1 HD	770.000.000
TL06M	HIT TV	618.000.000	RGE₁	La 1	770.000.000
TL06M	MegaStar	618.000.000	RGE₁	Clan	770.000.000
TL06M	CGTN-Español	618.000.000	RGE₁	24h	770.000.000
TL06M	Canal Galería	618.000.000	M P E₂	nova	778.000.000
TL06M	Business TV	618.000.000	M P E₂	neox	778.000.000
TL06M	8madrid	618.000.000	M P E₂	laSexta HD	778.000.000
RGE₂	tdp HD	634.000.000	M P E₂	laSexta	778.000.000
RGE₂	TEN	634.000.000	M P E₂	antena3 HD	778.000.000
RGE₂	DKISS	634.000.000	M P E₂	antena3	778.000.000
RGE₂	tdp	634.000.000			
RGE₂	Clan HD	634.000.000			









[II] Background: TV hijacking attacks



- East Coast USA 1986. At 12:32, HBO (Home Box Office) received its satellite signal from its operations center on Long Island in New York interrupted by a man who calls himself "Captain Midnight". The interruption occurred during a presentation by The Falcon and the Snowman.
- CHICAGO 1987 WGN (Channel 9) sportscast is hijacked at 9:14 pm on November 22. Someone wearing a Max Headroom mask and wearing a yellow blazer interrupted a recorded segment of the "Chicago Bears" for about 25 seconds. At 23:15 the broadcast of an episode of "Dr. Who" on the WTTW network was interrupted by the same character, this time with strange audio, an appearance of another person and a longer time in the air.
- Lebanon war 2006. During the Lebanon War of 2006, Israel overloaded the satellite broadcast of Al Manar TV of Hezbollah to broadcast anti-Hezbollah propaganda.



https://en.wikipedia.org/wiki/Broadcast_signal_intrusion



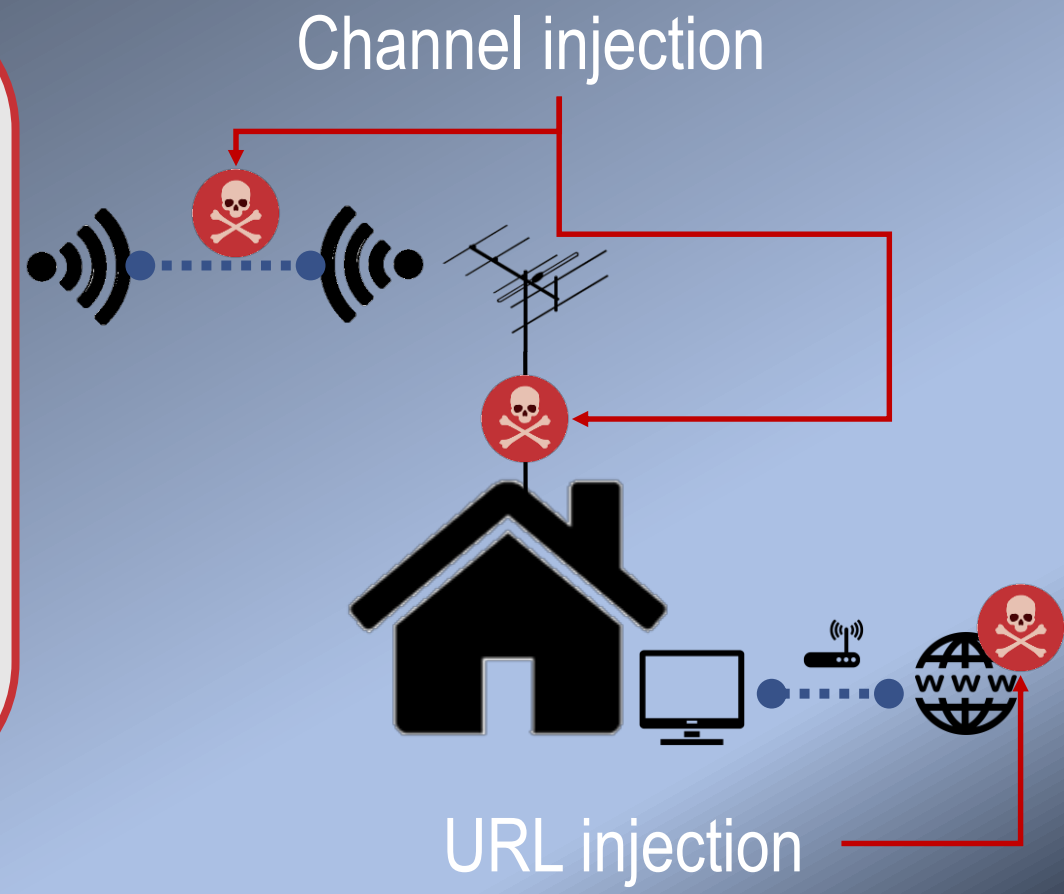
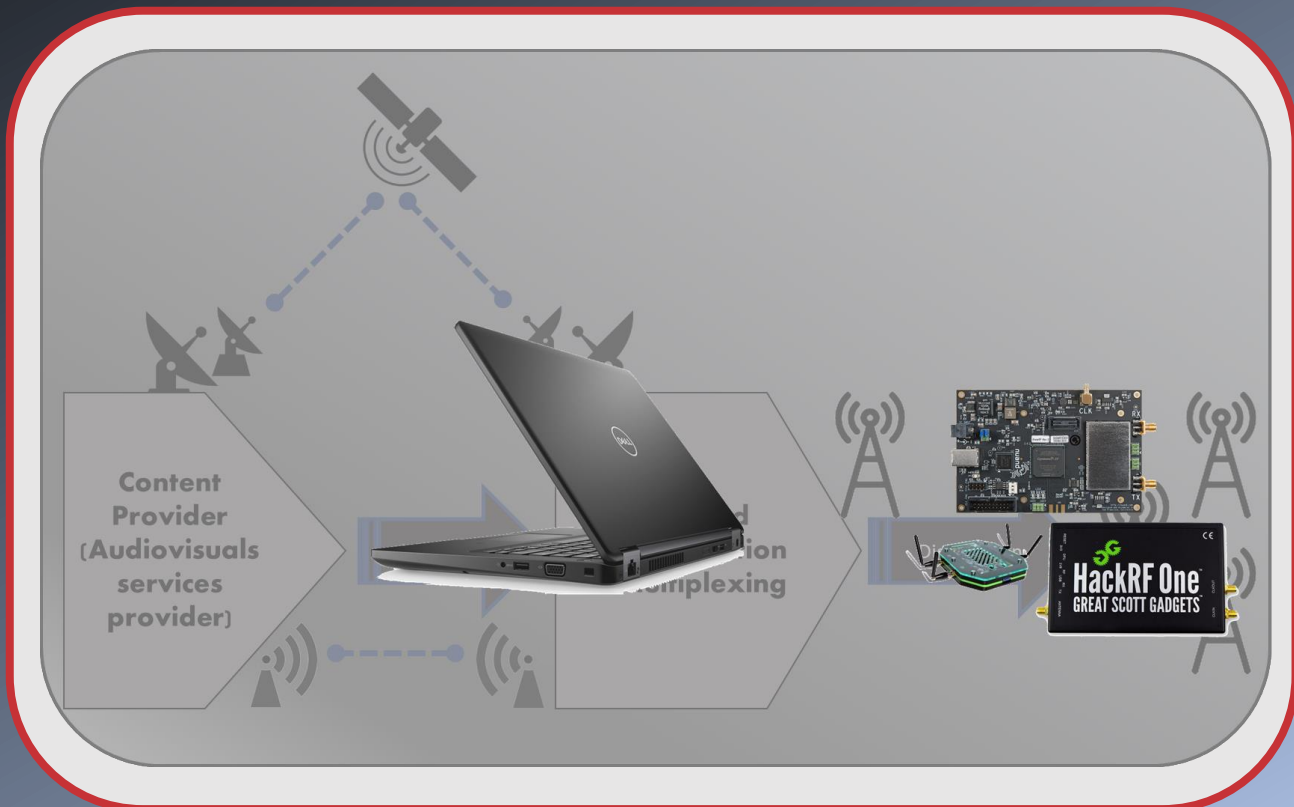
[II] Smart TV attacks state of the art



- June 2014 - Weeping Angel (CIA) - WikiLeaks. It shows exactly what an agent must do to turn a Samsung Smart TV into a microphone. Attack requires local access to the Smart TV.
- April 2015 - Yossef Oren and Angelos D. Keromytis "Attacking the Internet using Broadcast Digital Television". Theoretical study on the potential attacks on the HbbTV System.
- February 2017 - Rafael Scheel "Hacking a Smart TV". It presents two vulnerabilities to two Samsung Smart TV web browsers: Flash and Javascript, which it exploits by creating its own HbbTV application, broadcasting it through its own DVB-T channel. For this, it uses a low-cost proprietary device and an unpublished SW. In no case does it use SDR or OpenSource tools.



[II] DVB-T Channel Hijacking

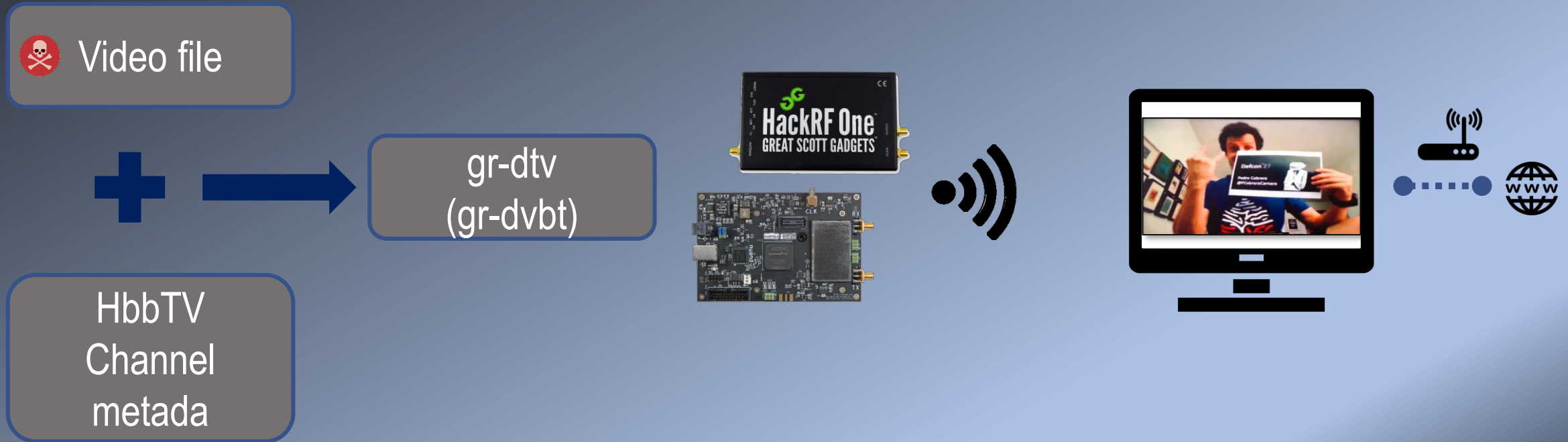




[II] DVB-T Channel Hijacking



Using the same frequency and channel metadata as in the original channel, we will transmit our video file using BladeRF, HackRF or any capable SDR supported by GNURadio:

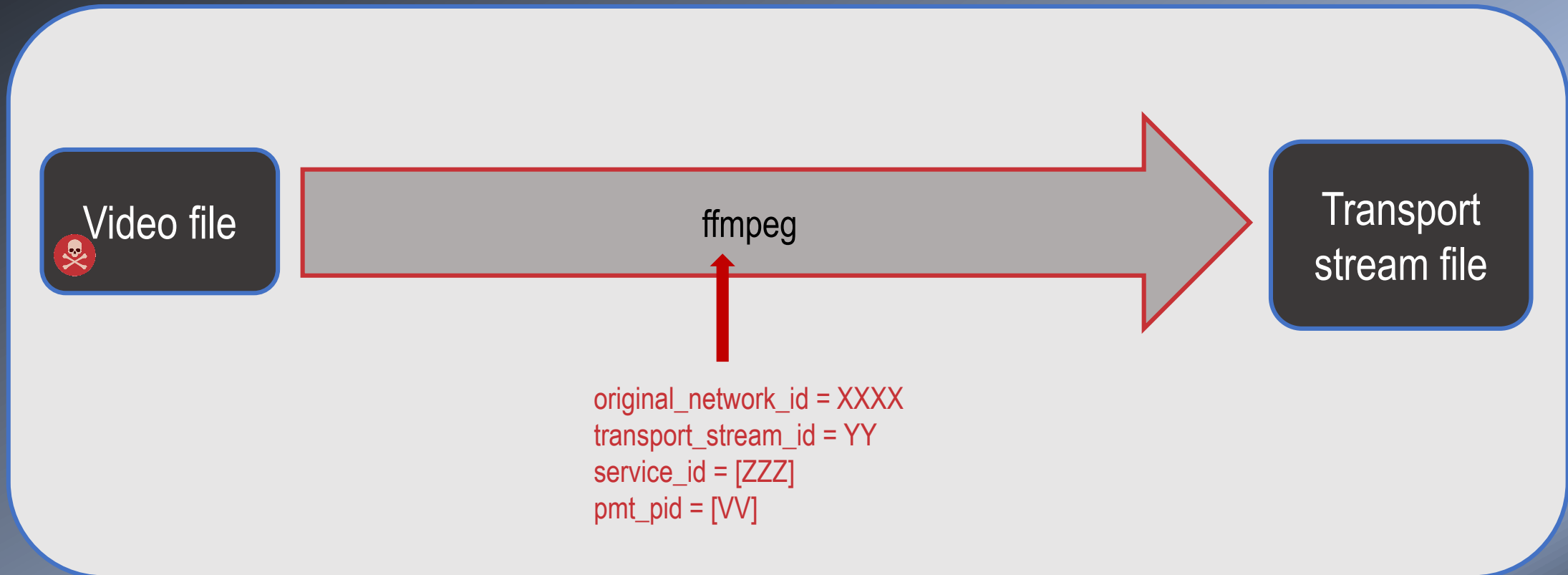




[II] DVB-T Channel Hijacking



We must generate a "Transport Stream" (TS file) with the same parameters of the legitimate channel and the new A/V content:

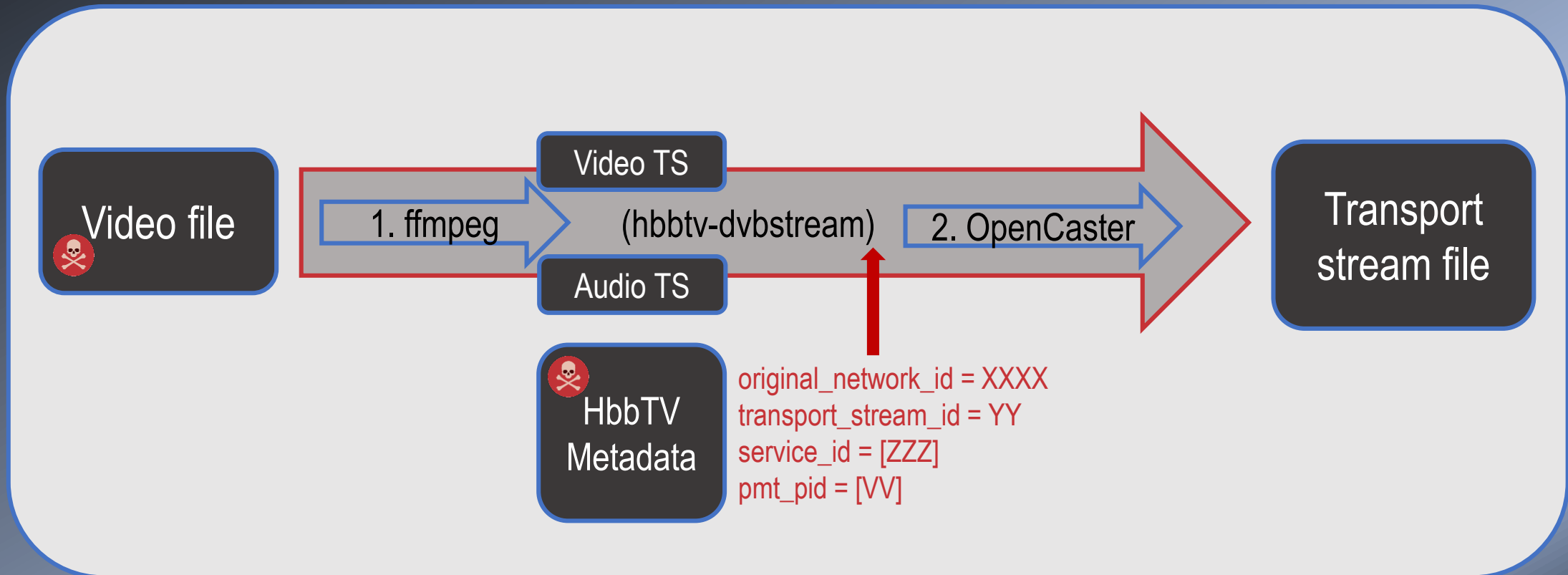




[II] DVB-T Channel Hijacking



We must generate a "Transport Stream" (TS file) with the same parameters of the legitimate channel and the new A/V content:





[II] DVB-T Channel Parameters



8madrid-1 - Kaffeine

Archivo Reproducir Lista de reproducción Televisión Preferencias Help

Buscar:

Canal Number

Canal	Number
+Galeria	1
24h	2
8madrid	3
8madrid-1	4
antena3	5
antena3 HD	6
BOM	7
BOTOPRO	8
BUENACO	9
Business TV	10
CADENA 1...	11
Canal Gal...	12
CGTN-Esp...	13
Clan	14
Déjate TV	15
Disney Ch...	16
DMAX	17
esRadio	18
GOL	19
Hit FM	20
Hit FM	21
Interecon...	22
Kiss FM	23
La 1	24
La 1 HD	25
La 2	26
La 2 HD	27
LA OTRA	28
LA OTRA	29

Inicio Reproducción Lista de reproducción Televisión

8madrid-1

Source: Terrestre

Frecuencia (MHz): 618

Ancho de banda: 8MHz

Modulación: 64-QAM

Ratio FEC: 2/3

Ratio FEC LP: NONE

Modo de transmisión: 8k

Intervalo de guardia: 1/4

Jerarquía: NONE

Id. de red: 8916

Id. de flujo de transporte: 39

Id. de servicio: 3901

Canal de audio: 257 (spa)

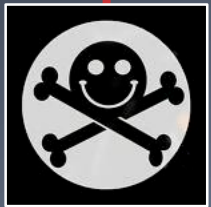
Mezclado: ☒

PMT PID: 272

PID de vídeo: 256

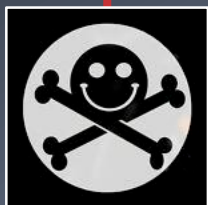
Cancel OK

spa desactivado 00:00:00





[II] DVB-T Channel Parameters



Linux command line:
dvbv5-scan (DVBv5 Tools)

```
Terminal - root@icarus: /home/pedro
Archivo  Editar  Ver  Terminal  Pestañas  Ayuda
[nova]
SERVICE_ID = 154
NETWORK_ID = 8916
TRANSPORT_ID = 15
VIDEO_PID = 601
AUDIO_PID = 603 604 605
PID_0c = 625
PID_06 = 611 202
PID_05 = 615
FREQUENCY = 778000000
MODULATION = QAM/64
BANDWIDTH_HZ = 8000000
INVERSION = AUTO
CODE_RATE_HP = 2/3
CODE_RATE_LP = 1/2
GUARD_INTERVAL = 1/4
TRANSMISSION_MODE = 8K
HIERARCHY = NONE
DELIVERY_SYSTEM = DVBT
root@icarus:/home/pedro#
```



[II] DVB-T Channel Hijacking



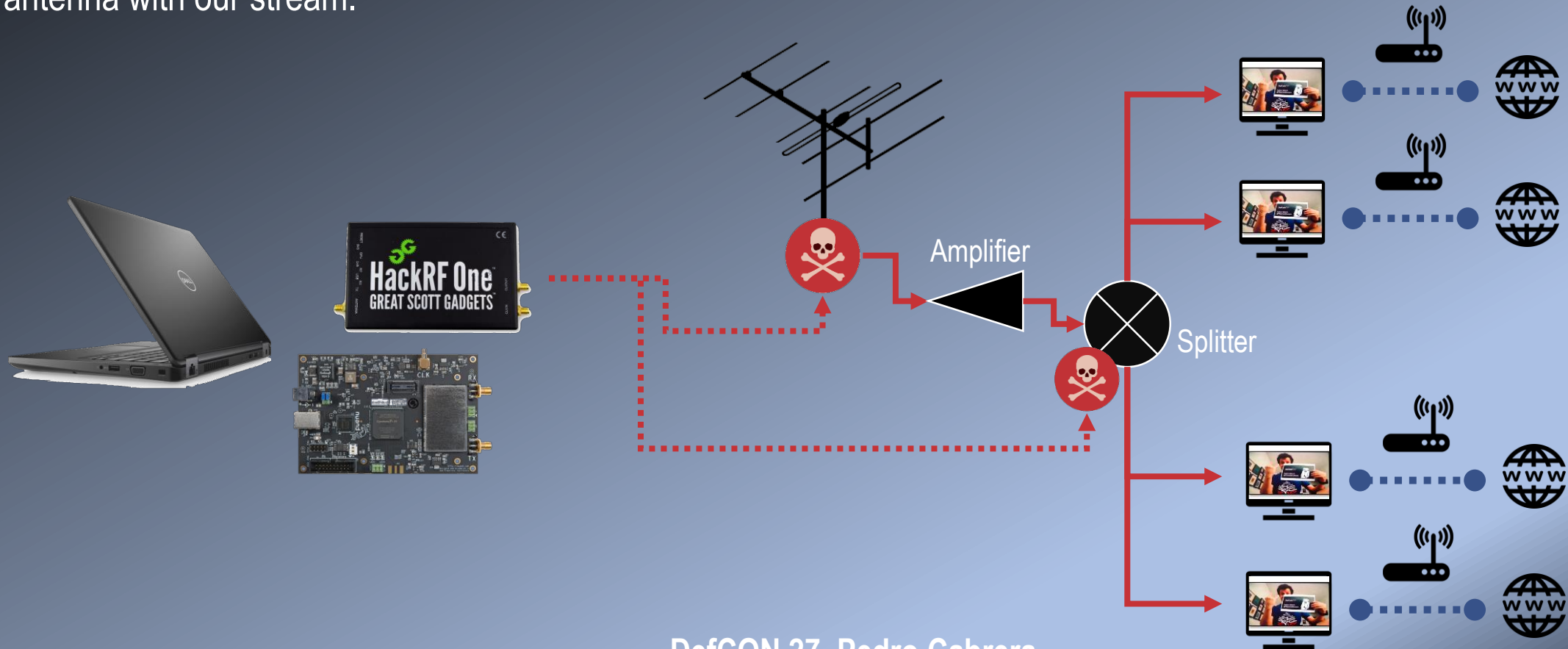
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[III] TV antenna facility attack

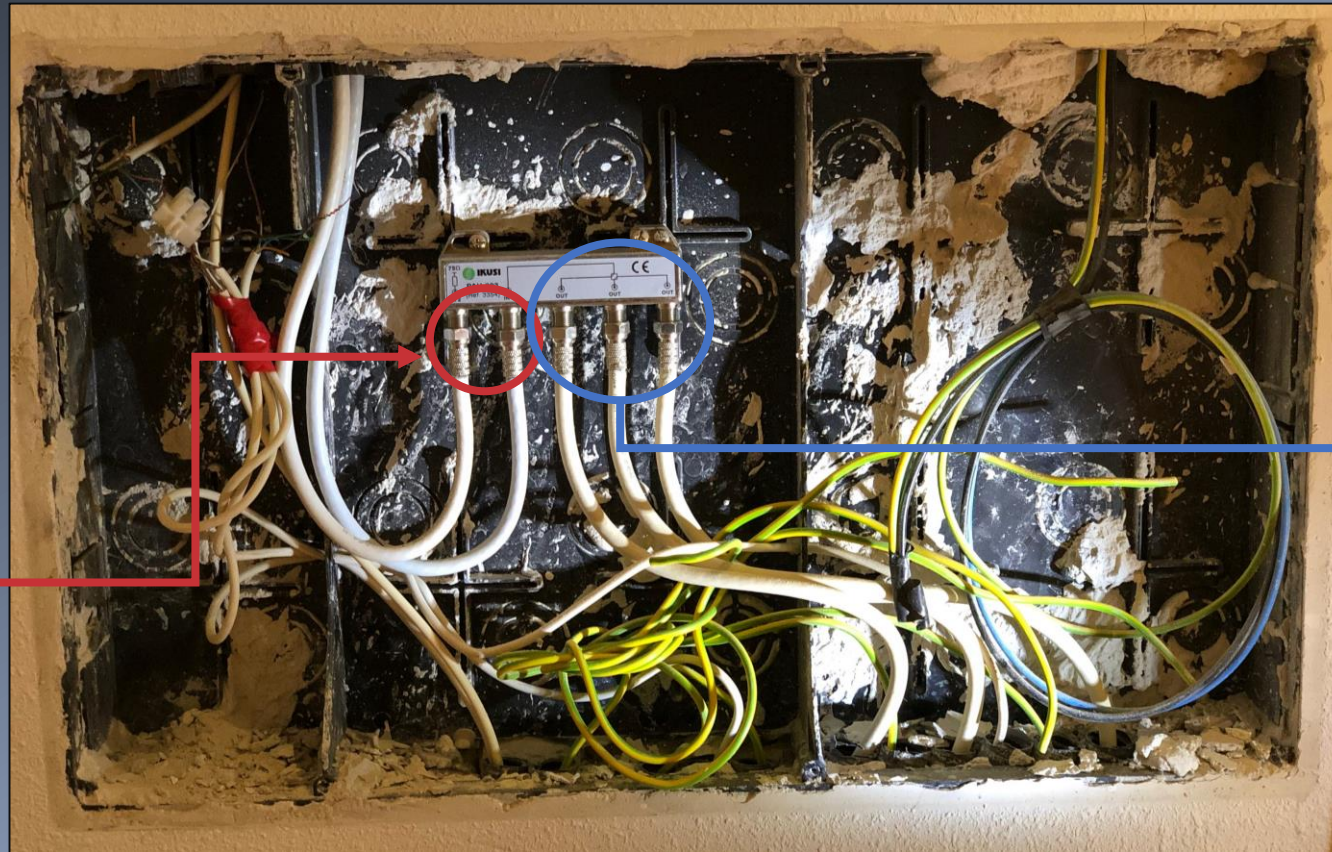


We can eliminate the radio phase by injecting our signal into the antenna facility, replacing the main TV stream from the antenna with our stream.





[III] TV antenna facility attack

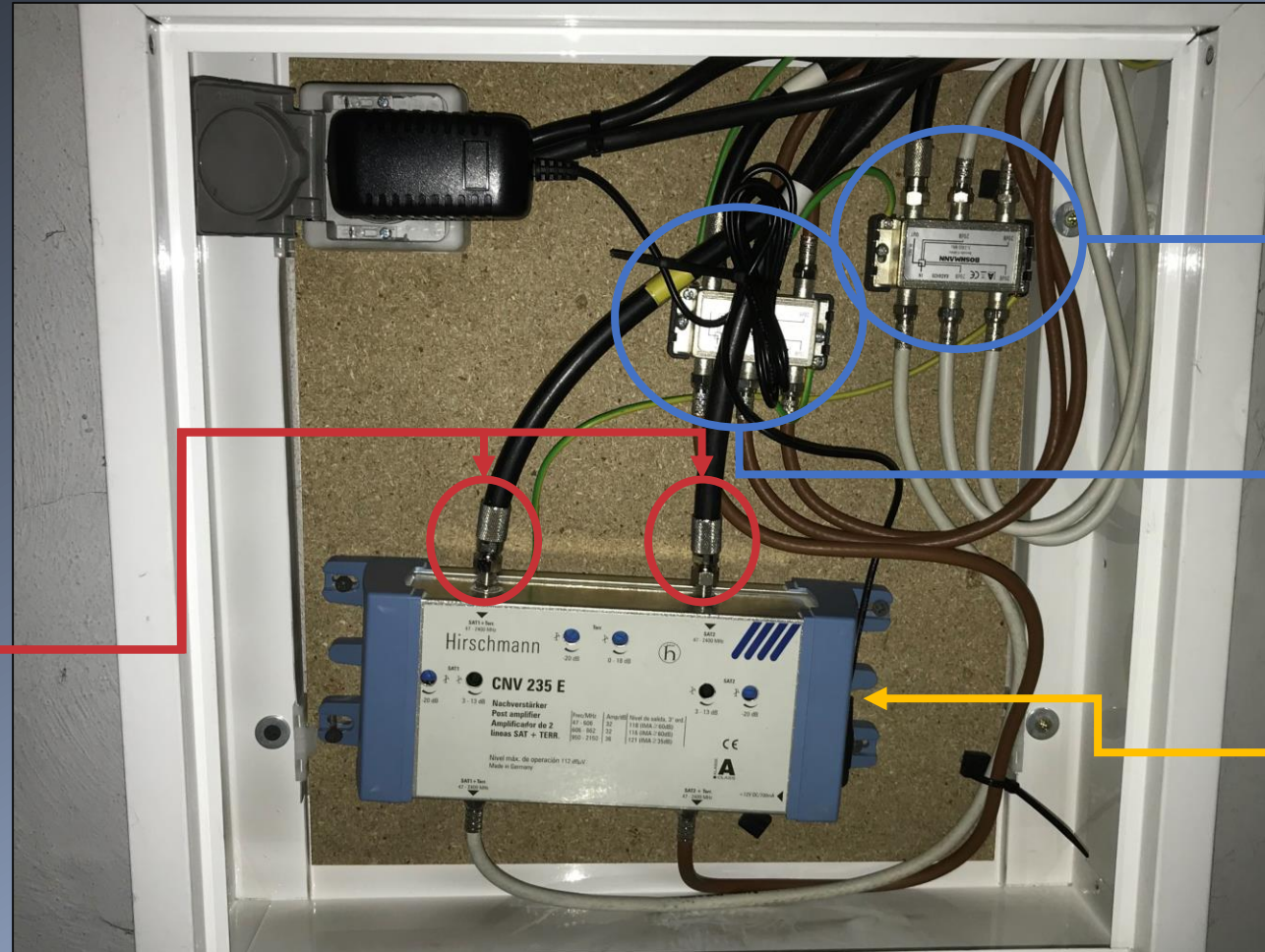


TV splitters (1/3)

TV antenna
facility



[III] TV antenna facility attack (II)



TV antenna
facility

TV splitters (1/4)

TV Amplifier



[III] TV antenna facility attack



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[III] Why miniaturization ?



Madrid > Fotografía y vídeo > Drones > Alquiler Dron Phantom 3 st

Alquiler de Dron Phantom 3 st



100.00 € por día

Carlos hace descuentos en función de la duración del alquiler.

Fin de semana (3 días): **250,00 €**

Semana (7 días): **450,00 €**

Desde el Hasta el

Reservar

The DJI Phantom 3 Standard is among the most popular drones of all time to date. Despite its design geared towards aerial photography, this does not stop it from being able to lift an additional payload. The DJI Phantom 3 Standard can lift around an additional 300 grams, or

<https://www.uavsystemsinternational.com/how-much-weight-can-a-drone-lift/>

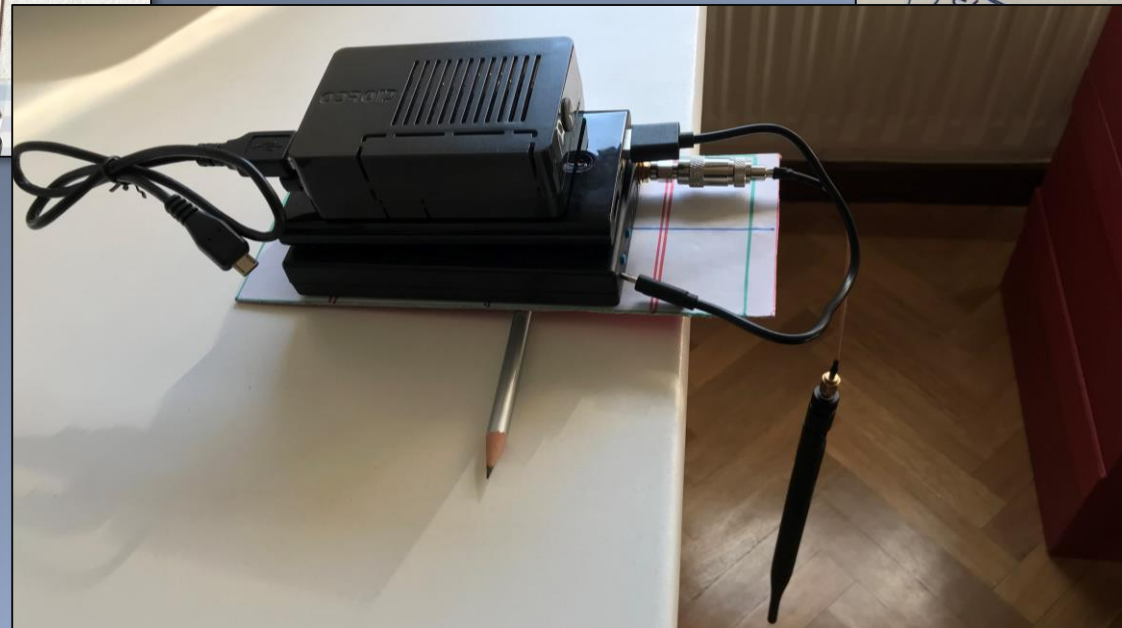
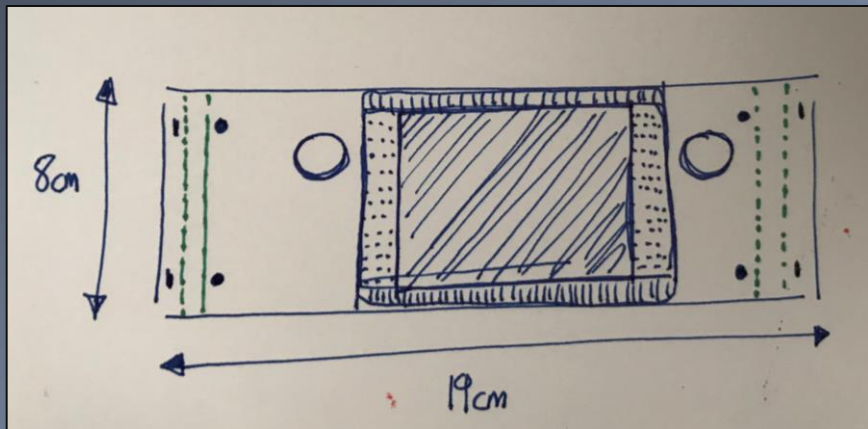
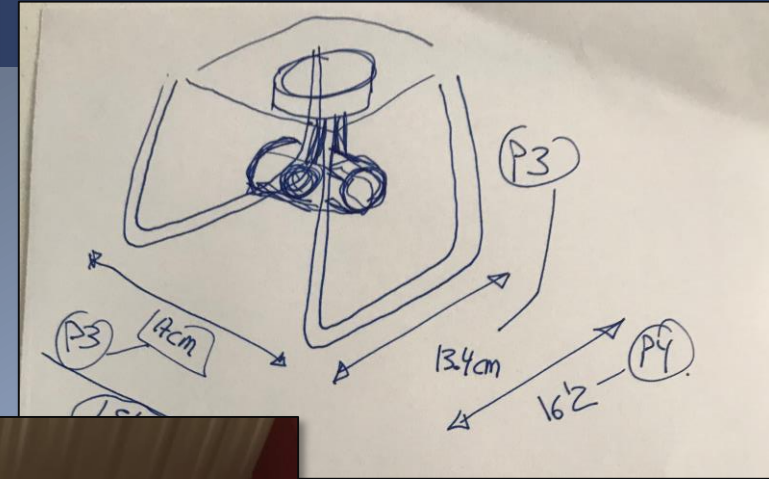
DefCON 27, Pedro Cabrera



[III] Miniaturization – Drone attacks



300 gr



GPD	480gr
BladeRF	170gr
HackRF	100gr
Bateria iPhone 10.000mA	280gr
Bateria Solar 24.000mA	350gr
Bateria NeoXeo 6.000mA	100gr
Odroid C2	68gr
Carcasa Odroid	32gr



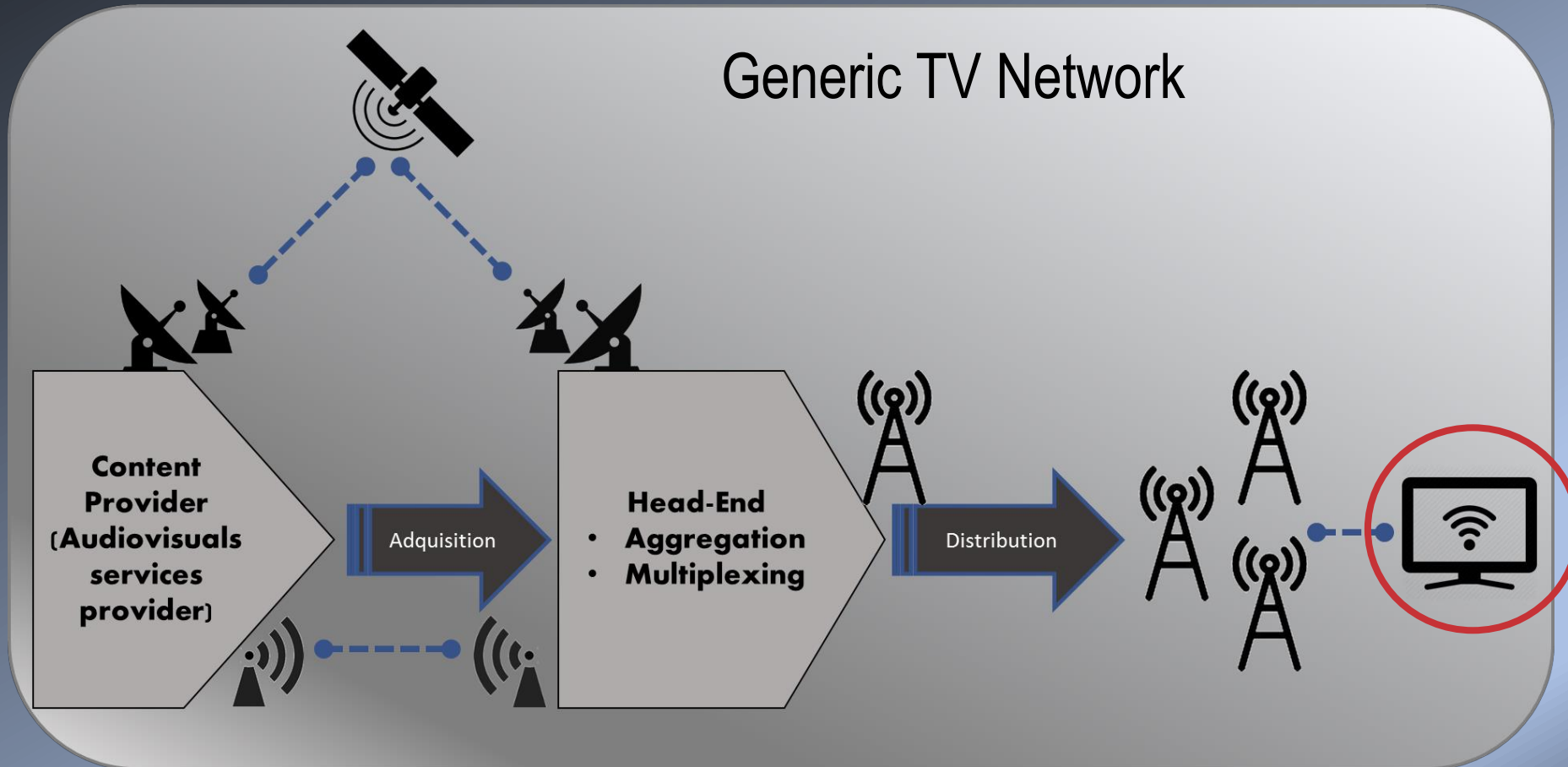
[III] Drone attack



rock'n roll FOREVER

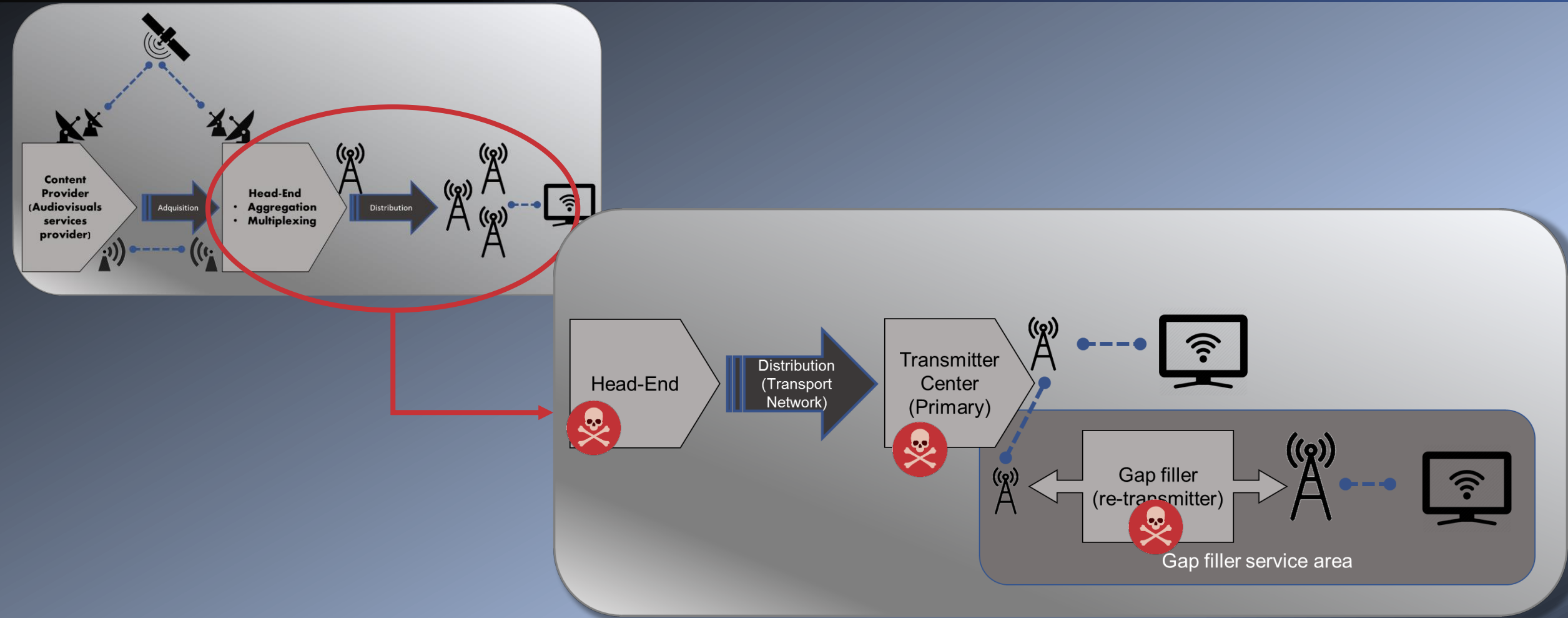


[III] DVB-T Channel Hijacking: Impact





[III] DVB-T Channel Hijacking: Impact

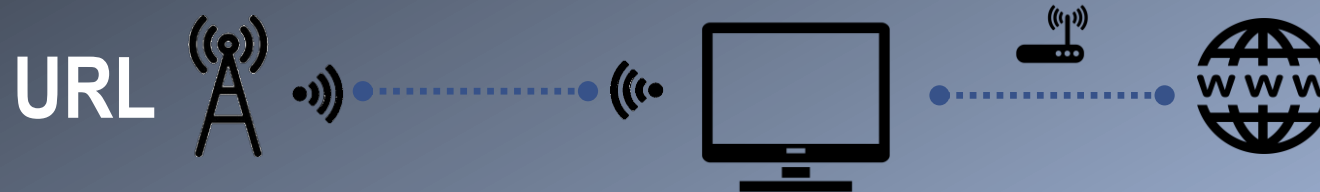




[IV] URL Injection attack



The HbbTV standard allows Smart TVs to send GET requests to the URL transmitted by the channel (station) every so often:



```
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?rnd=82575690 HTTP/1.1
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?rnd=82575690 HTTP/1.1
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?format=jsonp&rnd=87488938&callback=smartns_channel
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?format=jsonp&rnd=87488938&callback=smartns_channel
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?rnd=71453005 HTTP/1.1
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?rnd=71453005 HTTP/1.1
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?format=jsonp&rnd=43032479&callback=smartns_channel
GET /api/channels/atreseries.hbbtv.x.es.smartclip/channel_config?format=jsonp&rnd=43032479&callback=smartns_channel
GET /api/channels/atreseries.hbbtv.x.es.smartclip/current_adstatus?rnd=0B0A2F0A-5631-6559-0405-A65502695E8F.1825499
GET /api/channels/atreseries.hbbtv.x.es.smartclip/current_adstatus?rnd=0B0A2F0A-5631-6559-0405-A65502695E8F.1825499
```



[IV] URL Injection attack



Terminal - root@baleca /... *wlan0

92%

pedro 01 jul 12:25

*wlan0

Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http && http.request.method==GET

	Time	Source	Destination	Protocol	Length	Info
35	2019-07-01 10:24:58.542797308	192.168.0.14	81.93.189.220	HTTP	466	GET /latestversion HTTP/1.1
36	2019-07-01 10:24:58.542830138	192.168.0.16	81.93.189.220	HTTP	466	GET /latestversion HTTP/1.1
199	2019-07-01 10:25:02.928550272	192.168.0.14	31.170.190.122	HTTP	1030	GET /hbbtv/redbutton/launcher/app/v1.5.0-beta-10-6_4/inde...
200	2019-07-01 10:25:02.928605550	192.168.0.16	31.170.190.122	HTTP	1030	GET /hbbtv/redbutton/launcher/app/v1.5.0-beta-10-6_4/inde...
215	2019-07-01 10:25:09.360322354	192.168.0.14	77.209.227.57	HTTP	1014	GET /hbbtv/la1.html HTTP/1.1
216	2019-07-01 10:25:09.360337770	192.168.0.16	77.209.227.57	HTTP	1014	GET /hbbtv/la1.html HTTP/1.1
235	2019-07-01 10:25:09.535492326	192.168.0.14	185.103.39.27	HTTP	966	GET /hbbtv/redbutton/launcher/index.html?mode=ganchok26c...
236	2019-07-01 10:25:09.535525204	192.168.0.16	185.103.39.27	HTTP	966	GET /hbbtv/redbutton/launcher/index.html?mode=ganchok26c...
241	2019-07-01 10:25:09.603485597	192.168.0.14	185.103.39.27	HTTP	1030	GET /hbbtv/redbutton/launcher/app/v1.5.0-beta-10-6_4/inde...
242	2019-07-01 10:25:09.603524639	192.168.0.16	185.103.39.27	HTTP	1030	GET /hbbtv/redbutton/launcher/app/v1.5.0-beta-10-6_4/inde...
303	2019-07-01 10:25:09.718524227	192.168.0.14	8.247.214.124	HTTP	915	GET /css/rtve.2015.hbbtv/rtve.hbbtv.common/rtve.hbbtv.ba...
304	2019-07-01 10:25:09.718555223	192.168.0.16	8.247.214.124	HTTP	915	GET /css/rtve.2015.hbbtv/rtve.hbbtv.common/rtve.hbbtv.ba...
307	2019-07-01 10:25:09.720271630	192.168.0.14	8.247.214.124	HTTP	918	GET /css/rtve.2015.hbbtv/rtve.hbbtv.common/rtve.hbbtv.la...
308	2019-07-01 10:25:09.720279211	192.168.0.16	8.247.214.124	HTTP	918	GET /css/rtve.2015.hbbtv/rtve.hbbtv.common/rtve.hbbtv.la...
344	2019-07-01 10:25:09.720341006	192.168.0.14	8.247.214.124	HTTP	444	GET /css/rtve.2015.hbbtv/rtve.hbbtv.common/rtve.hbbtv.la...

Frame 35: 466 bytes on wire (3728 bits), 466 bytes captured (3728 bits) on interface 0
Ethernet II, Src: WistronN_b6:04:ca (68:02:b4:b6:04:ca), Dst: IntelCor_3d:3b:69 (a8:88:69:3d:3b:69)
Internet Protocol Version 4, Src: 192.168.0.14, Dst: 81.93.189.220
Transmission Control Protocol, Src Port: 53305 (53305), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 466
Hypertext Transfer Protocol

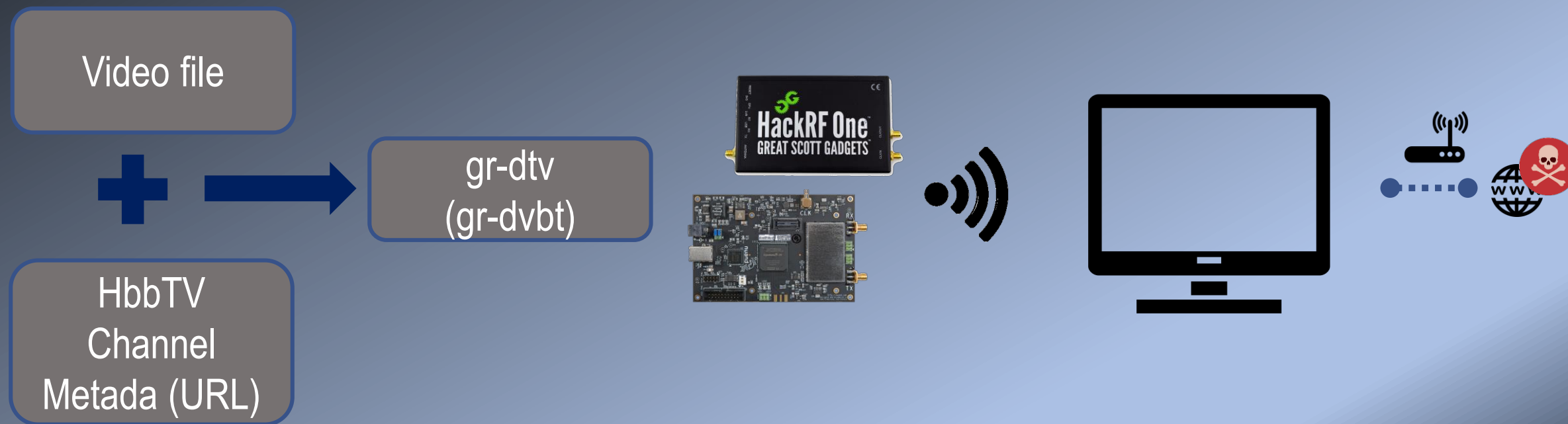
a0 88 69 3d 3b 69 00 02 b4 06 04 ca 08 00 45 00 ...1=;1...E.
01 c4 00 fe 40 08 40 06 68 46 c0 a8 00 0e 51 5d ...0.0. hF....Q]
bd dc 00 39 00 50 53 96 a8 49 4a b4 20 05 80 10 ...9.PS. .I].
04 02 a7 04 00 00 01 01 00 0a 00 01 cc 00 30 3b0;
3a 30 47 a5 64 30 3f 6c 01 7a 65 73 74 76 ff 790



[IV] URL Injection attack: Basic



We add the URL of our fake server in the HbbTV metadata: application name, base URL, web page, organizationId and applicationId





[IV] URL Injection attack: Video Replay



DVBv5 Tools
dvbsnoop

Channel video
& audio



gr-dtv
(gr-dvbt)

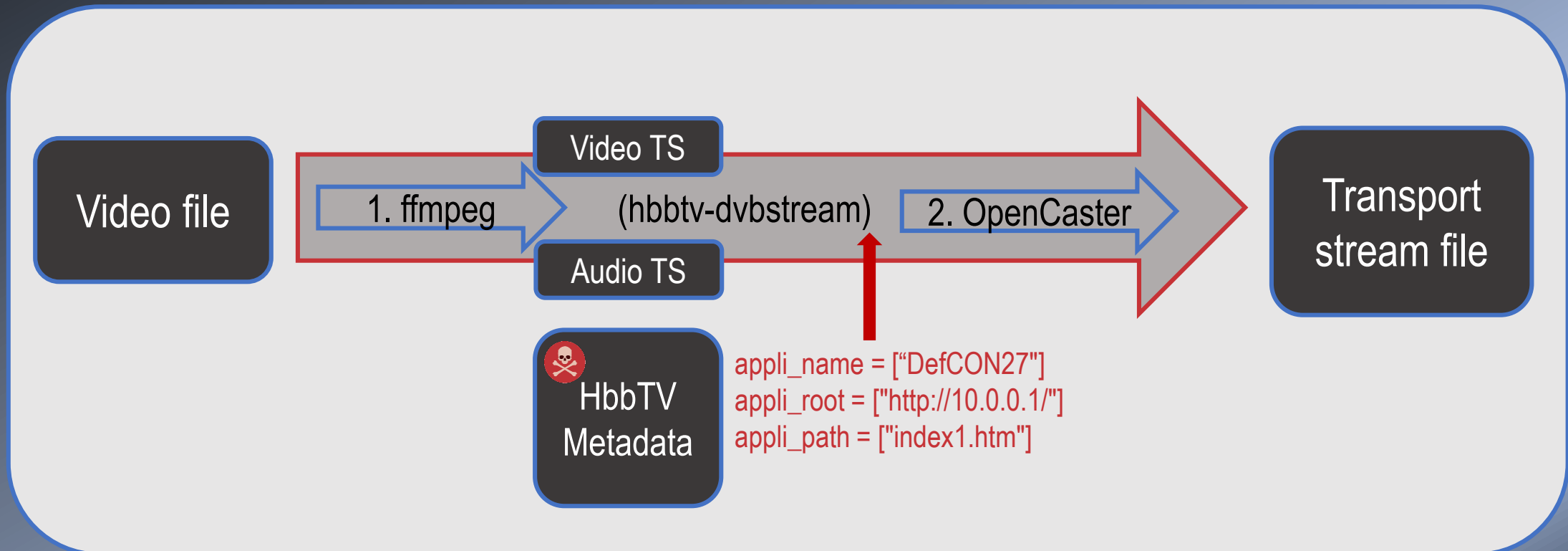
HbbTV
Channel
Metada (**URL**)





[IV] URL injection attack

We must generate a "Transport Stream" (TS file) with the same parameters of the legitimate channel and the new Application/URL content:





[IV] One SmartTV, two browsers



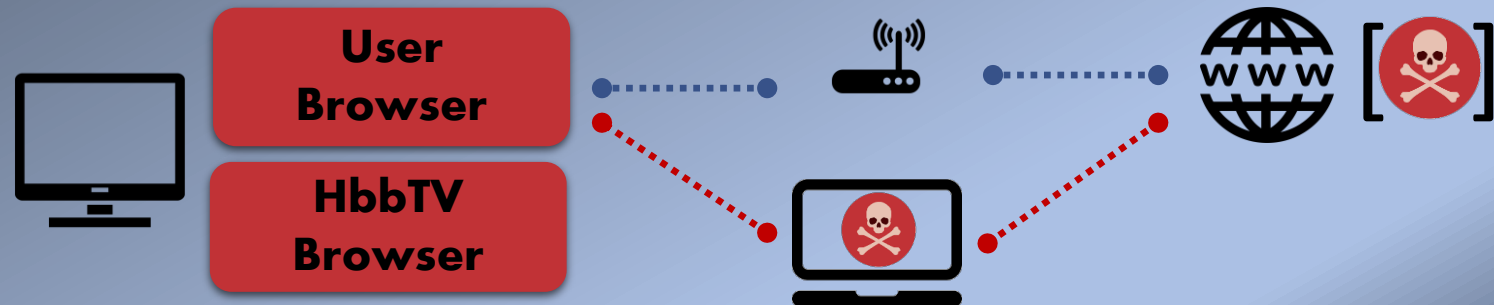
SDR URL injection attack

- HbbTV Browser (remote)



ARP Poison/DNS Hijacking URL injection attack

- HbbTV & User Browsers (requires WLAN access)





[IV] One SmartTV, two browsers



Samsung TV:

HbbTV/1.2.1 (+DRM+TVPLUS;Samsung;SmartTV2017;T-KTMDEUC-1106.2;;)

Mozilla/5.0 (SMART-TV; Linux; Tizen 3.0) AppleWebKit/537.36 (KHTML, like Gecko) SamsungBrowser/2.0
Chrome/47.0.2526.69 TV safari/537.36

Panasonic TV:

HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)

Mozilla/5.0 (X11; FreeBSD; U; Viera; es-ES) AppleWebKit/537.11 (KHTML, like Gecko) Viera/3.10.14
Chrome/23.0.1271.97 Safari/537.11



[IV] Smart (TV) scanning



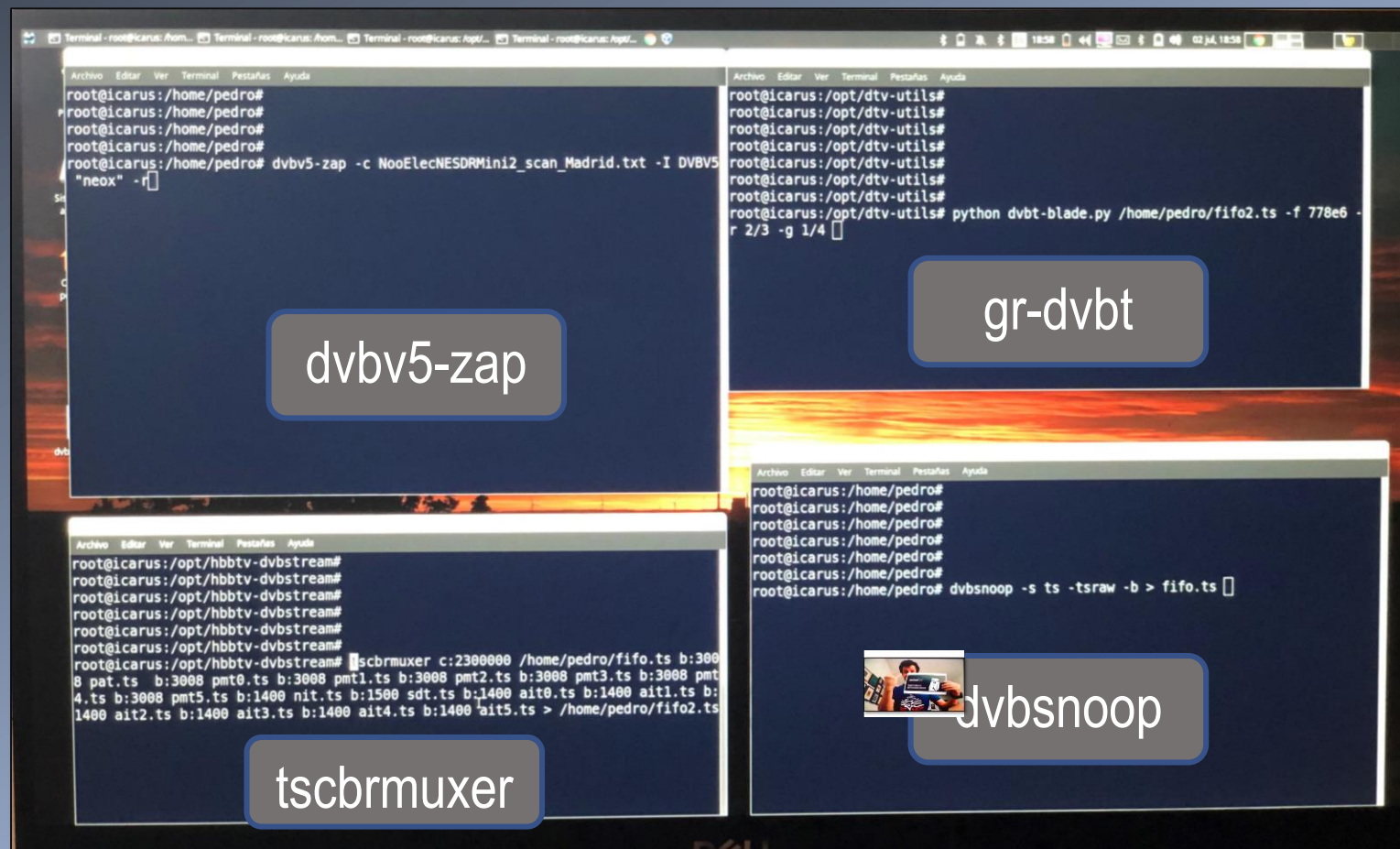
Apache Log files:

- Public IP address
- Models/Manufacturers (UA)
- DVB-T Channels/Audience analysis

```
Terminal - root@babieca:/var/log/apache2
Archivo Editar Ver Terminal Pestañas Ayuda
10.0.0.60 - - [20/Feb/2018:13:03:02 +0100] "GET /lasexta/app/scenes/Score.js HTTP/1.1" 200 794 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:03:02 +0100] "GET /lasexta/app/scenes/ShortcutToScene.js HTTP/1.1" 200 498 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:03:02 +0100] "GET /lasexta/resources/language/langConfig.js HTTP/1.1" 200 358 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:03:02 +0100] "GET /lasexta/resources/language/eng.js HTTP/1.1" 200 441 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:03:02 +0100] "GET /lasexta/app/javascripts/phaser.js HTTP/1.1" 200 432897 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/index.html HTTP/1.1" 200 806 "-" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/javascripts/head_min.js HTTP/1.1" 200 3849 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/loader.js HTTP/1.1" 200 811 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/stylesheets/styleSheet.css HTTP/1.1" 200 600 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/stylesheets/Loading.css HTTP/1.1" 200 420 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/stylesheets/MainMenu.css HTTP/1.1" 200 584 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/stylesheets/GameState.css HTTP/1.1" 200 726 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/stylesheets/Score.css HTTP/1.1" 200 416 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/javascripts/jquery.js HTTP/1.1" 200 72895 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/javascripts/state_machine.js HTTP/1.1" 200 2796 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/javascripts/srtObjects.js HTTP/1.1" 200 5240 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/common.js HTTP/1.1" 200 2906 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
10.0.0.60 - - [20/Feb/2018:13:04:40 +0100] "GET /lasexta/app/javascripts/soundPlayer.js HTTP/1.1" 200 9130 "http://hbbtv.atresmedia.com/lasexta/index.html" "HbbTV/1.2.1 (;Panasonic;VIERA 2014;3.101;6101-0003 0010-0000;)"
```

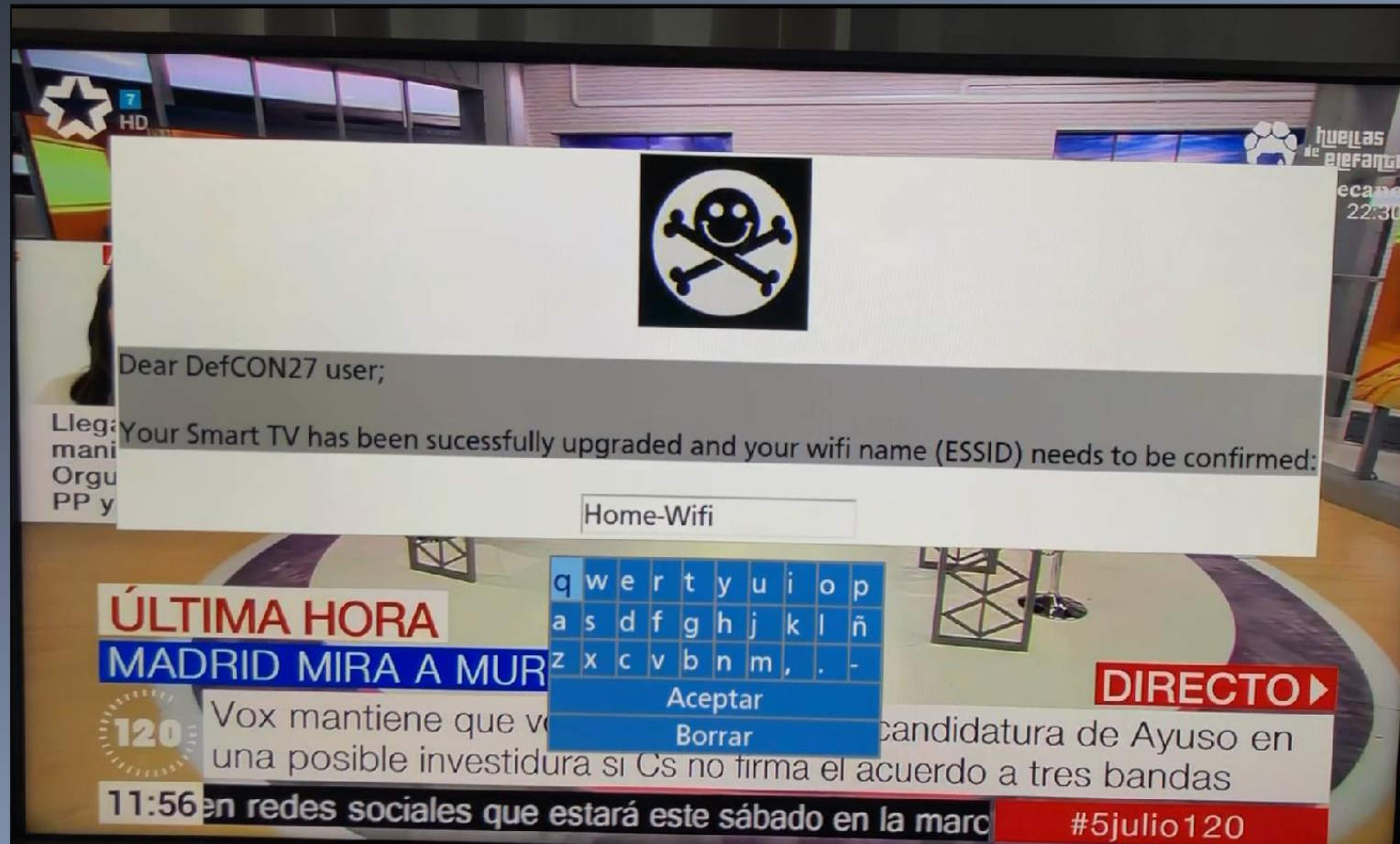



[IV] Video replay & URL injection attack



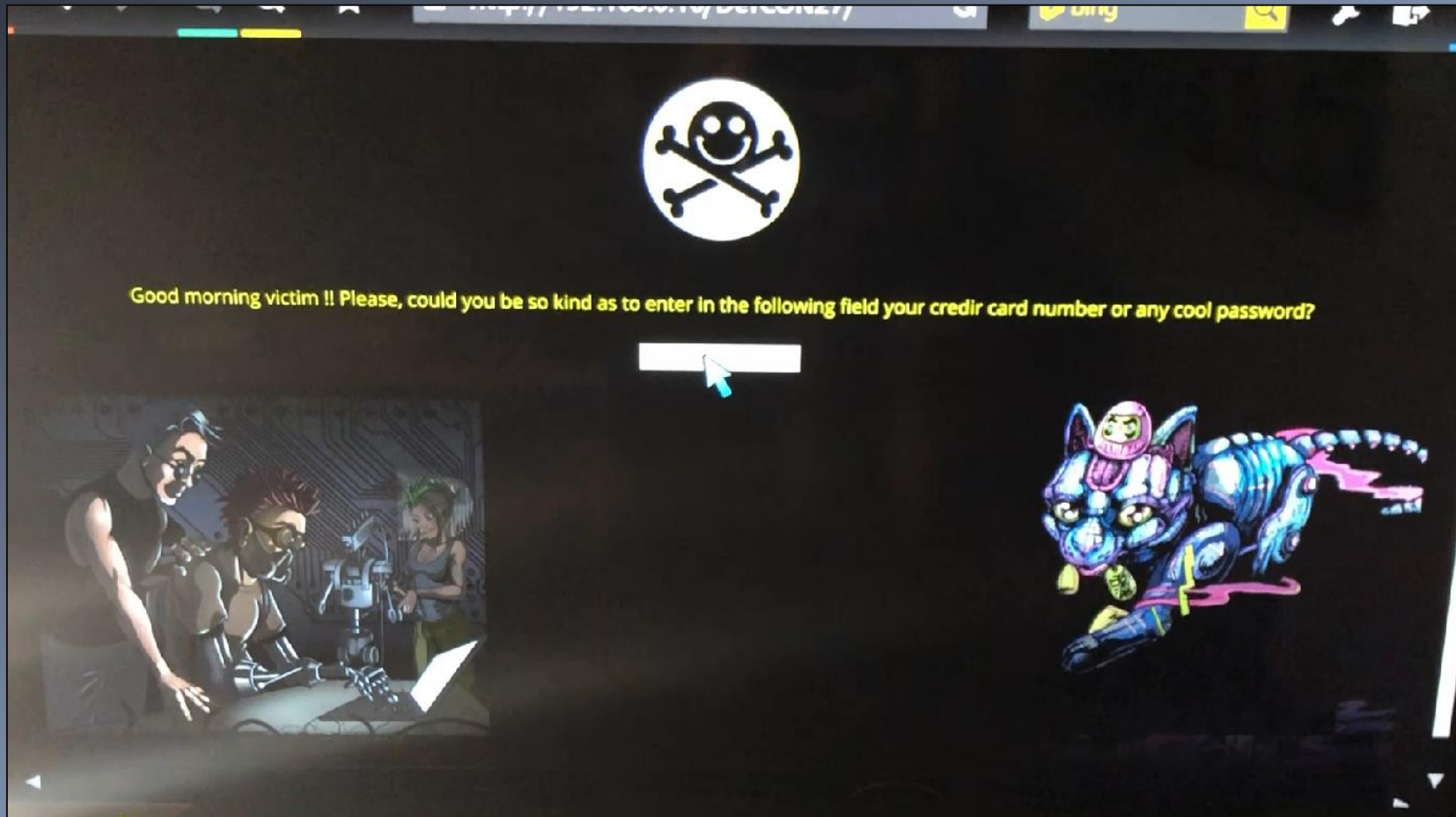


[IV] Social engineering (SE) attacks





[IV] Keylogger attack





[IV] Crypto Mining



Hackers Infect 50,000 Servers With Sophisticated Crypto Mining Malware

<https://www.coindesk.com/hackers-infect-50000-servers-with-sophisticated-crypto-mining-malware>

Cryptojacking: Hackers Just Want to Borrow your CPU

<https://medium.com/tebs-lab/cryptojacking-hackers-just-want-to-borrow-your-cpu-ebf769c28537>

Cryptocurrency HACKERS use YOUTUBE to target computers for bitcoin and ripple mining

HACKERS have been targeting users of YouTube to mine cryptocurrencies such as bitcoin by attacking computers through the video platform's advertising service, it has been reported.

<https://www.express.co.uk/finance/city/911278/cryptocurrency-hacking-bitcoin-ripple-ethereum-mining-youtube-adverts>

4K+ Websites Infected with Crypto-Miner after Tech Provider Hacked

<https://www.tripwire.com/state-of-security/latest-security-news/4k-websites-infected-with-crypto-miner-after-tech-provider-hacked/>

 **Scott Helme** ✓
@Scott_Helme

Ummm, so yeah, this is *bad*. I just had @phat_hobbit point out that @ICOnews has a cryptominer installed on their site... 🤖

♡ 912 3:46 PM - Feb 11, 2018

💬 881 people are talking about this



[IV] Crypto Mining attack





[IV] Hooking user browser



localhost:3000/ui/panel

Aplicaciones Evernote Google ING DIRECT

Hooked Browsers

- Online Browsers
 - www.youtube.com
 - 10.0.0.182
 - 10.0.0.182
 - 10.0.0.182
 - 10.0.0.182
- Offline Browsers
 - staticxx.facebook.com
 - 10.0.0.64
 - 10.0.0.64
 - 10.0.0.64

Getting Started Logs Current

Details Logs Commands Rider XssRays Ipec

Category: Browser (6 Items)

Browser Version: UNKNOWN

Browser UA String: Mozilla/5.0 (SMART-TV; Linux; Tizen 3.0) AppleWebKit/537.36 (KHTML, like Gecko) SamsungBrowser/1.0

Browser Language: es-ES

Browser Platform: Linux armv7l

Browser Plugins: Native Client

Window Size: Width: 392, Height: 220

Category: Browser Components (12 Items)

Flash: No

VBScript: No

PhoneGap: No

Google Gears: No

Web Sockets: Yes

Page URI: http://www.youtube.com/embed/Z8vDIDsh224?wmode=transparent

Metasploit Modules

Once Metasploit has been [configured](#) and launched, Metasploit modules are directly included in the BeEF command modules tree:

Module Tree

Metasploit

Metasploit (579)

- Firefox 5.0 - 15.0.1 __exposedProps__ XCS Code Execution
- Firefox 8/9 AttributeChildRemoved() Use-After-Free
- Firefox Proxy Prototype Privileged Javascript Injection
- Firefox XMLSerializer Use After Free
- Firefox location.QueryInterface() Code Execution
- Firefox nsSVGValue Out-of-Bounds Access Vulnerability
- Foxit Reader Plugin URL Processing Buffer Overflow
- Mozilla Firefox 3.6.16 mChannel Use-After-Free
- Mozilla Firefox Array.reduceRight() Integer Overflow
- Mozilla Firefox Bootstrapped Addon Social Engineering Code Execution
- Mozilla Firefox Interleaved document.write/appendChild Memory Corruption
- Mozilla Suite/Firefox compareTo() Code Execution
- msf_firefox_proto_crmrequest
- msf_firefox_proxy_prototype
- msf_firefox_queryinterface
- msf_firefox_xpi_bootstrapped_addon
- msf_foxit_reader_plugin_url_bof
- msf_mozilla_attrchildremoved

Browser Version: UNKNOWN

Browser UA String: Mozilla/5.0 (SMART-TV; Linux; Tizen 3.0) AppleWebKit/537.36 (KHTML, like Gecko) SamsungBrowser/1.0

Browser Language: es-ES

Browser Platform: Linux armv7l

Browser Plugins: Native Client

Window Size: Width: 392, Height: 220



[IV] User browser attack





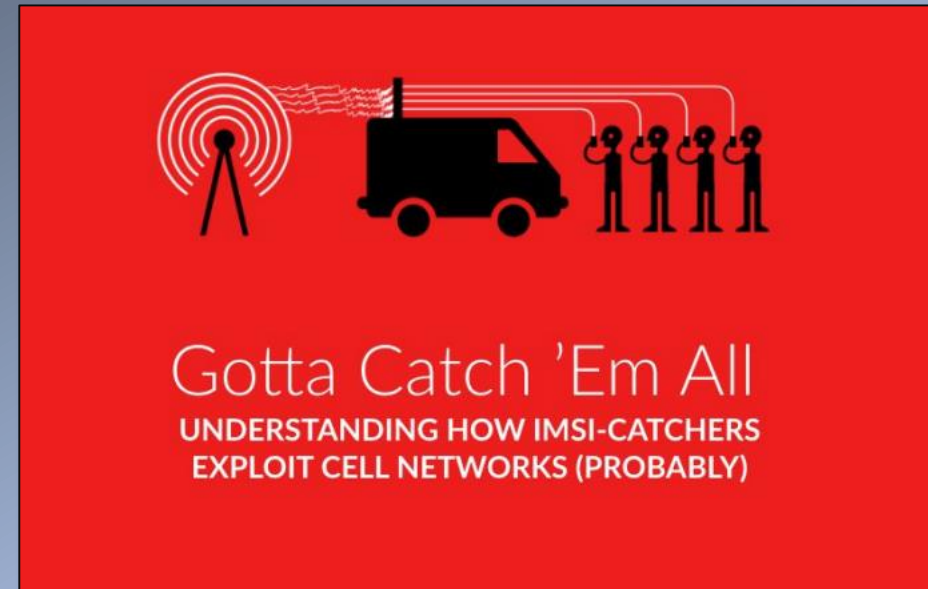
[V] Conclusions



Mass GPS Spoofing Attack in Black Sea?



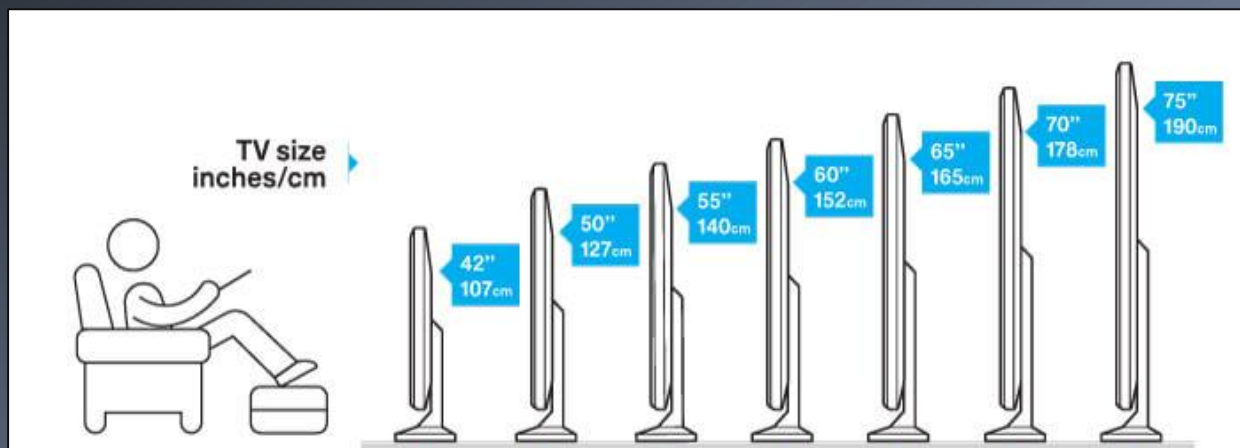
<https://maritime-executive.com/editorials/mass-gps-spoofing-attack-in-black-sea>



https://www.eff.org/files/2019/07/09/whitepaper_imsicatchers_eff_0.pdf

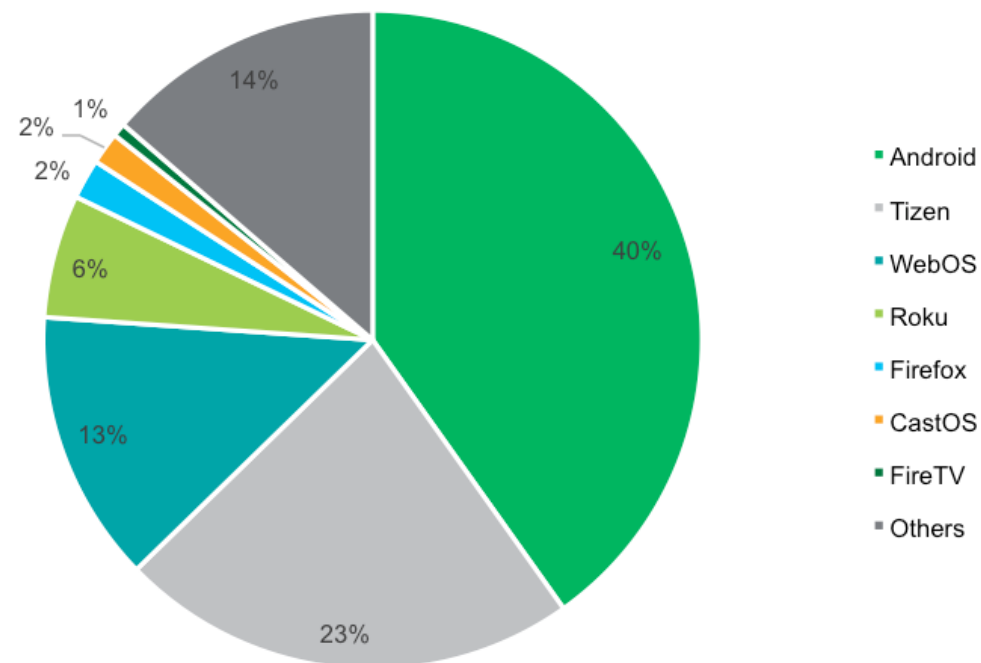


[V] Conclusions



<https://www.choice.com.au/electronics-and-technology/home-entertainment/tvs-and-projectors/buying-guides/tvs>

2018 Smart TV Operating System Share



Source: IHS Markit TV Sets Intelligence Service Premium

© 2018 IHS Markit

<https://voicebot.ai/2018/07/19/smart-tv-market-share-to-rise-to-70-in-2018-driven-by-streaming-services-alexa-and-google-assistant/>

Thank You

Gonzalo Manera
Pepe Cámara
Alvaro Castellanos
Luis Bernal (aka n0p)

github.com/pcabreraacamara/DC27

2019 August, DefCON 27